



Fast response MPPT charger for the Técnico Solar Boat

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Introduction to the Research in
Electrical and Computer Engineering

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Abstract

Solar-powered vehicles, particularly boats, need effective energy conversion in rapidly changing conditions. The Maximum Power Point Tracker (MPPT) charger plays a critical role in energy conversion, ensuring the panels operate at their Maximum Power Point (MPP). This work focuses on the design, simulation and testing of a fast response MPPT charger for Técnico Solar Boat (TSB).

The prototype was designed to charge a lithium-ion battery of 43.2 V nominal voltage. The proposed charger is based on a synchronous boost converter operating at 100 kHz controlled by a Microcontroller Unit (MCU).

Different MPPT techniques were analyzed to identify a suitable solution for the dynamic operating conditions of the vessel. Balancing tracking speed and accuracy led to the implementation of a Perturb and Observe (P&O) algorithm. The converter and Photovoltaic (PV) system were simulated in both LTspice and Matlab to validate the component selection and algorithm parameters. A custom Printed Circuit Board (PCB) was designed, manufactured and soldered by hand.

Experimental results show that the implemented MPPT algorithm can respond rapidly to changes in irradiance or temperature, with the measured tracking response being suitable for the environmental variations identified. The prototype achieved a measured efficiency of up to 96%. Beyond performance, a computer application was also developed for configuration and data analysis. The resulting system provides a dedicated, configurable, and low-cost MPPT charger for TSB and a platform for future improvements.

Keywords

Keywords: MPPT, MPPT topology, DC-DC Converter, Solar energy, P&O, Solar Boat

Resumo

Veículos alimentados a energia solar, em particular embarcações, necessitam de uma conversão de energia eficaz em condições climáticas de rápida variação. O carregador MPPT desempenha um papel importante nesta conversão, garantindo que os painéis operam no seu MPP. Este trabalho centra-se no projeto, simulação e ensaio de um carregador MPPT de resposta rápida para o TSB.

O protótipo foi projetado para carregar uma bateria de íões de lítio com tensão nominal de 43,2 V. O carregador proposto baseia-se num conversor boost síncrono a operar a 100 kHz, controlado por um MCU.

Foram analisadas diferentes técnicas de MPPT de modo a identificar uma solução adequada às condições de operação dinâmicas da embarcação. O compromisso entre velocidade de seguimento e precisão conduziu à implementação de um algoritmo P&O. O conversor e o sistema fotovoltaico foram simulados em LTspice e em Matlab para validar a seleção de componentes e os parâmetros do algoritmo. Foi projetada, fabricada e soldada manualmente uma PCB dedicada.

Os resultados experimentais mostram que o algoritmo de MPPT implementado responde rapidamente a variações de irradiância ou de temperatura, com um tempo de seguimento adequado às variações ambientais identificadas. O protótipo atingiu uma eficiência medida até 96%. Para além do desempenho, foi também desenvolvida uma aplicação para configuração e análise de dados. O sistema resultante constitui um carregador MPPT dedicado, configurável e de baixo custo para o TSB, bem como uma plataforma para desenvolvimentos futuros.

Palavras Chave

Palavras chave: MPPT, topologia MPPT, conversor DC-DC, energia solar, P&O, barco solar

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Acronyms

AC	Alternating current
ASIC	Application-Specific Integrated Circuit
BMS	Battery Management System
CAN	Controller Area Network
CC	Constant Current
CCM	Continuous Conduction Mode
CV	Constant Voltage
DC	Direct Current
DCM	Discontinuous Conduction Mode
ESL	Equivalent Series Inductance
ESR	Equivalent Series Resistance
FPGA	Field Programmable Gate Array
GUI	Graphical User Interface
IC	Integrated Circuit
IncCond	Incremental Conductance
MCU	Microcontroller Unit
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracker
LUT	Look-Up Table

OC	Over Current
OTS	Off the shelf
P&O	Perturb and Observe
PCB	Printed Circuit Board
PV	Photovoltaic
R&D	Research and Development
SG01	São Gabriel 01
SR03	São Rafael 03
STC	Standard Testing Conditions
TSB	Técnico Solar Boat

1

Introduction

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This thesis presents the design, simulation, implementation and experimental validation of a custom Maximum Power Point Tracker (MPPT) charger for the photovoltaic system of Técnico Solar Boat (TSB). The charger is intended to extract the maximum available power from the team's solar panels under the rapidly changing irradiance of a marine environment, while charging the low voltage battery of São Gabriel 01 (SG01) and providing telemetry through a Controller Area Network (CAN) interface.

This chapter introduces the context of the work, the motivation for developing an in-house MPPT, the project objectives, and the organization of the remainder of this document.

1.1 Motivation

With environmental pollution intensifying worldwide, the transition to cleaner energy sources such as solar power has become increasingly urgent. In 1985, only 20.82% of the energy produced in the world came from renewable energies. These numbers have been rising every year, and in 2024, have reached 31.92% [11]. In 2024, 6.9% of the energy produced in the world come from solar panels, and in Portugal this number rises to 14.5%, demonstrating the importance and the impact of solar energy [11]. Solar energy still plays a minuscule role, and it is listed behind the other sources of energy in terms of its contribution to meeting the world's energy demand. However, solar energy is becoming more relevant, with the cost of solar panels dropping significantly in the last decade.

Compared with other renewable sources, photovoltaic energy is advantageous owing to its availability, simplicity, low maintenance, environmental friendliness and reliability. More recently, it is becoming more relevant in the automotive industry, with solar powered cars, boats, and robots [12]. The CO₂ emissions of the automotive sector are one of the main contributors to global pollution. In 2018, 29% of total CO₂ emissions in the EU came from the transport sector, with 4% coming from the maritime sector alone [13]. These challenges have directly motivated the development of the TSB project.

In 2015, TSB was created with the goal of designing and building solar powered boats to compete in international competitions while also contributing to the innovation in yachting industry. Since then, the project has grown, and several vessels have been built. It began with the construction of the first solar prototype, São Rafael 01, which had a lot of room for improvement and was followed by São Rafael 02 and 03. The team then decided to approach the hydrogen energy and autonomous driving, and therefore São Miguel 01 and 02 and São Pedro 01 were built. More recently, a vessel was built to combine all the technologies used before and was named SG01.

All these prototypes used solar energy to maximize their range and efficiency. In the first years the energy produced was small, and the whole system was built using commercial Off the shelf (OTS) components. However since the beginning of the TSB project, the spirit of innovation and the desired to develop all the systems in house has been present. In previous years, the team has built its own

batteries, telemetry systems, flight controllers, fuel cell controllers, etc. Some other more complex systems were design during master thesis. Until today a Battery Management System (BMS), dual motor system, Marine Propulsion System, a MPPT and more.

In 2020, the team started to build its own solar panels for São Rafael 03. To increase performance and the amount of information acquired by the solar panels' operation, in the same year, surge the first attempt to do a MPPT. This was the fist step to achieve a fully functional product. The result was a working MPPT that was never implemented on any vessel due to constantly burning components which was not reliable. Additionally, the project is outdated in terms of components and firmware, lacking configurability.

The MPPT charger is a fundamental subsystem in photovoltaic energy systems, responsible for maximizing the power extracted from solar panels by ensuring continuous operation at the Maximum Power Point (MPP). A MPPT charger is typically a power converter that continuously monitors the voltage and current produced by the photovoltaic modules and adjusts their electrical operating point accordingly. By dynamically regulating these operating parameters, the MPPT compensates for variations in environmental conditions such as solar irradiance and temperature, which cause the MPP to shift over time. As a result, the photovoltaic system can operate with improved efficiency and deliver maximum available power under changing conditions [14].

In the context TSB project, a solar system is specially important to increase the autonomy of our vessels. In 2023/24 season we were able to consumption of São Rafael 03 (SR03) from 1.5 kW to 300 w by using solar energy Decreasing 80 % of his consumption. With a costume made MPPT, we believe that we could reach even better performances wile also increasing the data acquired about the system performance that would serve to improve the system.

Now in 2025/26 season, the relevancy of the solar system decreased with our new vessel SG01, since it uses around 20 KW of constant power. So the main power source will be hydrogen. However, solar energy will continue to be used in this vessel to power the auxiliary systems being a reliable green source that will keep the boat powered in all situations.

1.2 Objectives

This project aims to design and implement a MPPT charger for the solar panels used in TSB project. The MPPT charger will convert the energy produced by the solar panel as efficiently as possible with the use of a quality DC-DC converter and the implementation of MPP tracking algorithms.

The main objectives of this project are:

- Study and understand the operation of solar panels and MPPT techniques;
- Choose the most suitable DC-DC topology for the system;

- Implement a fast response and accurate control algorithm;
- Design and simulate the hardware and software;
- Implement the system in hardware and develop a Printed Circuit Board (PCB);
- Test and validate the performance of the designed circuit;
- Ensure the safety and reliability of the MPPT charger for its integration in the TSB project;
- Provide performance data of the solar panel and the MPPT through a CAN communication interface;
- Develop a practical solution to change the prototype configuration.

By achieving these objectives, the project will contribute to a more efficient solar system, maximizing the energy received from the solar panels and increasing the available data for performance tracking of the MPPT and solar panel. This data will be used to later improve the energy efficiency of the MPPT and take conclusions about the manufacture quality of the solar panels build by TSB project. A cheap, efficient and versatile MPPT will also mean future savings to the team, since it could be used in a variety of vessels.

In the context of TSB, such complex projects with high scientific rigor, that involve simulations, PCB design, laboratory tests and field tests are only possible in master thesis. So, for TSB, the main objective is to increase the team knowledge about this subject by increasing the documentation about it. Also, due to a drastic event of another team, which burn all their MPPTs during competition, the security of this system is the main concern.

1.3 Outline

This document is organized into seven chapters:

- **Chapter 1** introduces the context of the work, presenting the motivation behind the development of a custom MPPT charger for the TSB project. The objectives of the project are defined, and the structure of the document is outlined;
- **Chapter 2** describes the electrical architecture of SG01, including the high and low voltage systems, the photovoltaic installation, and the batteries used in this work. The contribution of the solar system to vessel autonomy is analysed, and the time scales of environmental variations are estimated in order to define the required tracking speed of the MPPT;
- **Chapter 3** reviews the State-of-the-Art of photovoltaic systems and MPPT technology. It covers the operating principles of solar panels, the most common MPPT algorithms, non isolated DC-DC converter topologies, lithium-ion battery charging techniques, and commercially available solutions;
- **Chapter 4** presents the implementation of the proposed charger. The design of the synchronous boost converter, the selection of the processing unit and sensors, and the firmware architecture are described, including the charging state machine and communication interfaces;
- **Chapter 5** reports the simulations used to validate the converter design and the MPPT algorithm. A photovoltaic cell model and the boost converter are simulated in LTspice, while an analytical power-loss model is developed in MATLAB to estimate the efficiency and the contribution of each component. Additionally a using MATLAB/simulink, the algorithm speed and accuracy was tested ;
- **Chapter 6** presents the experimental validation of the prototype. Three hardware iterations are assembled and tested in the laboratory, covering sensing accuracy, converter efficiency, thermal behavior, MPPT tracking performance, and field tests;
- **Chapter 7** summarizes the main conclusions of this work and identifies directions for future development.

2

System

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This chapter presents the electrical architecture of São Gabriel 01 (SG01), providing the necessary context for the development and integration of the proposed Maximum Power Point Tracker (MPPT) charger. The main subsystems relevant to solar energy conversion are presented, such as the installed solar panels and battery. The impact of the Photovoltaic (PV) system is analyzed as well as the environment changes.

2.1 Overview

SG01 is the most recent vessel built by TSB, which was designed to compete in Sea Lab category of Monaco energy boat challenge. To compete in this category a vessel with 8 x 2.5 m was designed creating new challenges for the team. This vessel was designed to use both hydrogen and solar energy, using hydro foils to hydroplane which highly increases efficiency, by reducing the drag. Additionally, it will use actuators and sensors to drive autonomously to compete in AI class of Monaco Energy Boat Challenge.

Since it is a larger and heavier boat than the previous ones built by TSB it needs more power than the others. So a motor with 130 KW peak power was chosen. This much power is only possible to provide with high voltages, so for the first time in TSB history this vessel have a High Voltage system (300-600 V) and a Low Voltage system (30-50 V).

The High Voltage system is responsible for the drive train, powering a motor. Since the energy consumed by this motor is much larger than solar panels can ever achieve, the boat is designed to be powered by hydrogen, however it is not implemented yet due to budget constraints.

As for the low voltage system, it is designed to power all auxiliary system (dashboard, telemetry, flight control, electrical string, throttle, etc.). Some of this auxiliary systems are critical for the normal use of the vessel, for example if the dashboard turns off, the pilot will not know the state of the high voltage system (temperature, voltage, state of charge, etc.). Or even more critical, if the throttle does not work, the boat can not electrically move.

The auxiliary systems are always changing and growing, so it is difficult to estimate the power consumption of this system. For this work it is considered 1000 W of continuous consumption. It is not a perfect estimation because the power consumption will depend on the driving mode (autonomous driving, hydroplaning or normal drive) which can create different power profiles due to the actuators. Additionally, the cooling system (which is planed to be moved to the High Voltage system) have high power consumption that is not considered in this estimation.

To help to maintain the low voltage system always powered, it is used solar panels which are connected to several MPPTs. In the future, there are plans to add a bidirectional DC-DC converter between the low voltage battery and the high voltage battery, enabling power transfer in both directions.

For better understanding of the system architecture see Figure 2.1 which represent highly simplified diagram of the overall system.

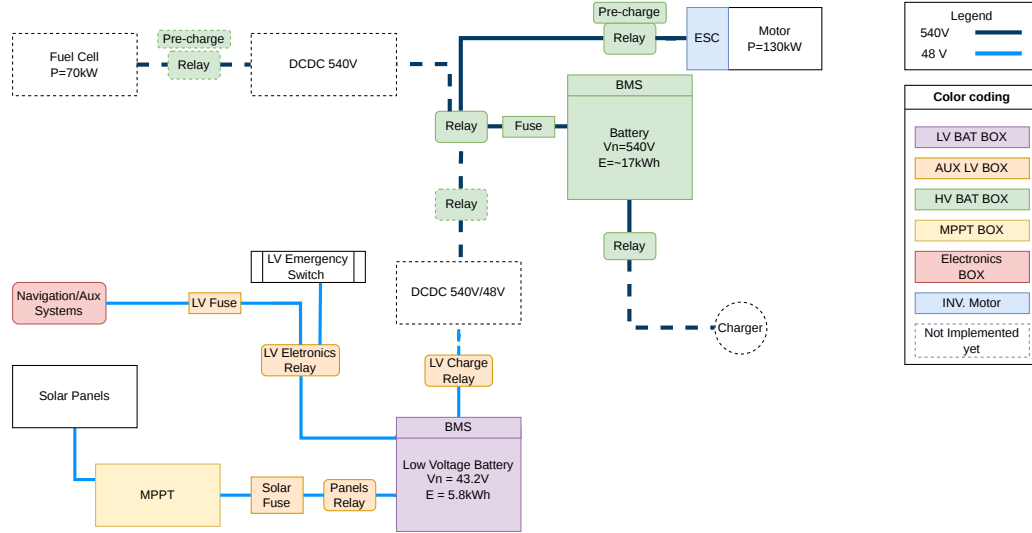


Figure 2.1: Simplified electrical schematic of SG01.

2.2 PV system

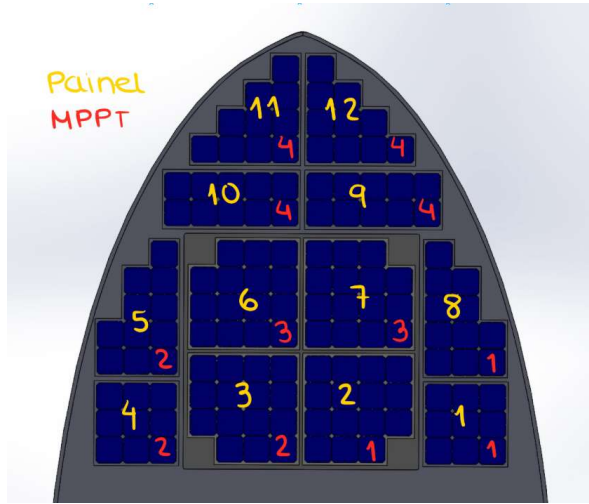
The photovoltaic system of SG01 was designed to maximize the energy harvested from the available deck area while maintaining reliable operation under the varying environmental conditions encountered in marine applications.

The current installation consists of 140 Maseon Gen 3 (Me3) photovoltaic cells distributed among ten custom-manufactured solar panels. These panels are connected to four independent MPPTs, as illustrated in Figure 2.2a. The distribution of panels among the MPPTs was carefully selected to reduce the effects of partial shading, temperature gradients, and non-uniform solar irradiance across the vessel surface. By electrically separating regions with different operating conditions, each MPPT can independently track the optimal operating point of its associated panels, increasing the overall energy yield.

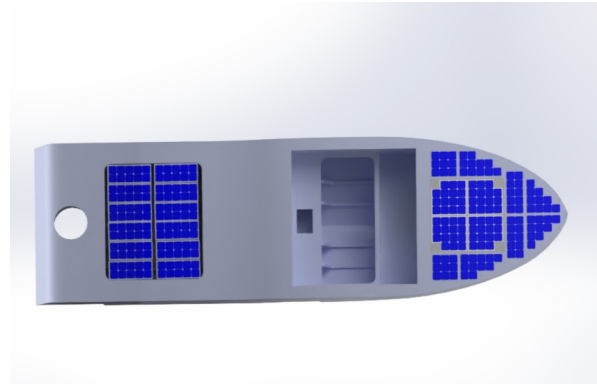
Each PV cell has an active area of approximately 155.06 cm^2 , resulting in a total photovoltaic area of 2.171 m^2 .

Assuming Standard Testing Conditions (STC) (irradiance of 1000 W/m^2 AND 25°C), the photovoltaic array is capable of delivering approximately 526.4 W of peak power (theoretical value).

The PV energy is extracted by four commercial Genasun MPPTs, which are grouped in a dedicated MPPT box. The MPPTs convert the variable voltage produced by the solar panels into a stable charging



(a) Distribution of Solar Panels and MPPT on SG01.



(b) Render of São Gabriel 01 with additional solar panels.

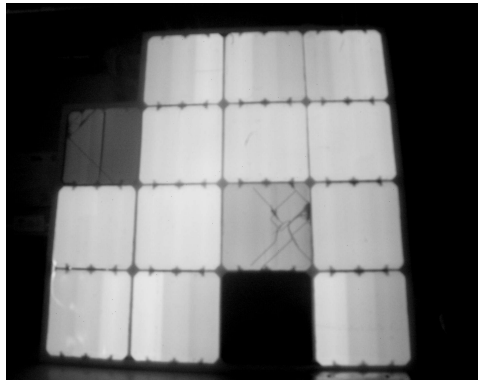
Figure 2.2: Solar panels installation on SG01.

voltage suitable for the low voltage battery system. This architecture allows each photovoltaic group to operate close to its Maximum Power Point (MPP) independently, improving energy extraction under non uniform illumination conditions.

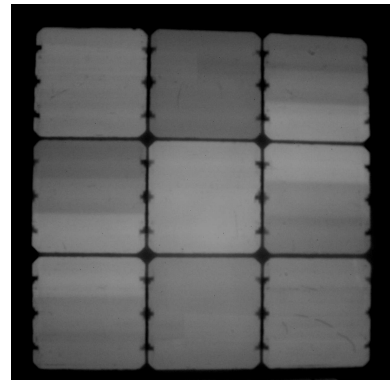
To ensure high performance and shape configurability, the photovoltaic modules were manufactured in house using materials selected specifically for marine operation. This process begins by soldering all cells in a module together and then a lamination is performed using vacuum and heat blacks.

Prior to installation, all panels underwent electroluminescence testing to detect manufacturing defects such as microcracks, broken cells, or interconnection failures, Figure 2.3. These tests were performed before integration into the vessel to ensure that only panels with acceptable electrical and structural integrity were installed. This quality control procedure reduces the likelihood of premature degradation and enables comparison before and after competitions.

Although the current photovoltaic installation already provides a significant contribution to the low-voltage energy budget, additional usable deck area remains available. Figure 2.2b presents a conceptual rendering with supplementary solar panels. Preliminary estimates indicate that the total photovoltaic power could be increased to approximately 977 W, almost doubling the currently installed capacity. Such an upgrade would substantially reduce battery charging times and significantly increase the autonomy of the vessel.



(a) Worst results.



(b) Best results.

Figure 2.3: Electroluminescence test results.

2.3 Battery

In the past years we have been producing our batteries achieving a lower cost and highly customizable solution. In SG01 we chose lithium-ion cells due to its ease of use and quick assembly. The assembly is done by the members. It begins by placing the cells on plastic holder to increase structural strength, then using a spot welder, the series and parallel connections are made with nickel tabs (Figure 2.4). In parallel, some thermistors are placed in different locations of the battery to ensuring a precise a robust temperature reading. In the end, the battery is placed in a case with a Battery Management System (BMS) and all the necessary connections.

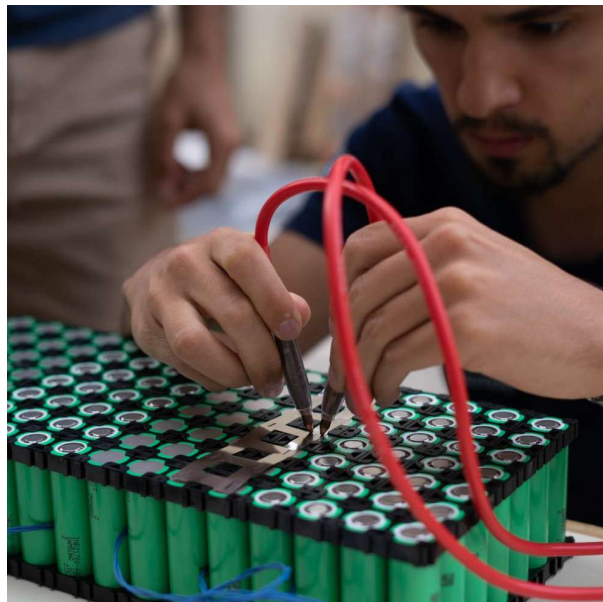


Figure 2.4: Low voltage battery assembly.

The current low voltage battery uses Samsung INR21700-50G cells in a 12s28p configuration. This

configuration results in the specifications written in Table 2.1.

Table 2.1: SG01 and São Rafael 03 (SR03) battery specs.

Parameter	Unit	SG01 Battery	SR03 Battery	NAVICTUS Battery
Max Voltage	V	50.4	50.4	50.4
Nominal Voltage	V	43.2	43.2	43.56
Min Voltage	V	30	30	30
Configuration	–	12s28p	12s7p	12s1p
Number of cells	–	336	84	12
Energy Capacity	Wh	5866.56	1512	223.9
Capacity	Ah	135.8	35	5.140
Max charge current	A	135.8	42	5.3
Cutoff charge current	A	3.395	1.75	0.133
Max discharge current	A	271.6	175	15.9
Weight	kg	25.2	?	1
Internal resistance	mΩ	6	24	192

SG01 battery was designed for a different vessel which needed a substantial higher capacity. The battery is too heavy, and overall overdimensioned.

For a more realistic impact analysis, SR03 battery will be used. This is a much smaller battery with a reasonable capacity for this vessel. It uses Samsung INR21700-50s cells in a 12s7p configuration. This configuration results in the specifications also written in Table 2.1.

Additionally, to reduce the time of the charging test another battery is used. This battery was provided by a startup called NAVICTUS and is a 12s1p with INR21700-53G cells from Samsung.

2.4 PV System Impact on Autonomy

With SG01 battery (5844.56 Wh) and the maximum power that the solar panels can currently provide (526.4 W), the minimum time to fully charge the battery is 11h and 7min (5844.56 Wh / 526.4 W). This is a large time to charge a battery. There are two solutions to this problem: increase the power provided by the solar panels or decrease the battery capacity.

Let's consider the case where we reduced the battery capacity of 1512 Wh which is the capacity of our solar boat (SR03) battery. With this change alone the battery is fully charged in 2h and 53min, which is a much more reasonable time. Additionally, if we increase the power provided by the solar panels to 977 W, the battery will be fully charged in 1h and 33min, which is an excellent time to charge a battery.

The charging times are a good to evaluate the PV system, but the main goal is to maximize the time on water before full battery discharge. For that lets consider 1000 W of consumption. Considering the original battery it takes 5 h 51 min to fully discharge without solar panels and 12 h 21min with solar panels and approximately 10 days with the additional solar panels. As previously said, this battery is overdimensioned and will be changed in the future.

To get a better idea of the impact of the PV system lets consider the battery of SR03. This battery takes 1 h and 30 min to fully discharge without solar panels and 3 h 12 min with solar panels and approximately 2 days 18 h with the additional solar panels.

By reducing the battery capacity and increase the solar panel area, the battery would never need external charging. In another point of view, the low voltage system can drain more power from the battery.

2.5 Environmental changes

In engineering, the term “fast”, used in the title of this thesis, needs to be associated with a time scale. A process can only be considered fast when compared with another process with a longer characteristic time. Therefore, this section investigates the characteristic time scales associated with variations in the MPP caused by environmental changes, with particular emphasis on the oscillatory motion of the vessel and its implications for the required speed of the MPPT algorithm.

When analyzing MPP variations, there two main factors: temperature and irradiance. Temperature variations are generally slower than variations in irradiance. Although rapid temperature changes can occur, for example when water comes into contact with the photovoltaic panels, such events are considered unusual during normal operation and are therefore not considered in this analysis. Irradiation on the other hand, can vary more rapidly as the angle of the solar panels changed with the vessel angle, particularly roll and pitch.

The motion of a vessel subjected to waves can be described as a forced dynamic system. The vessel has characteristic natural frequencies associated with its different modes of motion, including roll, pitch and heave. The response to wave excitation depends on the relationship between the excitation frequency and the corresponding natural frequency of the vessel. In particular, when the excitation frequency approaches a natural frequency, the response amplitude can increase significantly due to resonance. The vessel does not, however, necessarily oscillate at its natural frequency when subjected to excitation at another frequency, the steady state response occurs primarily at the excitation frequency, while the natural frequency determines the dynamic response and its amplitude.

Considering a simplified undamped free oscillation system, the natural frequency of a vessel can be calculated using the natural period Formula 2.1 where I is the mass moment of inertia and C is the restoring stiffness [15].

$$T = 2\pi\sqrt{\frac{I}{C}} \quad (2.1)$$

When applied to pitch and roll, the restoring stiffness is given by the mass (m), gravity (g) and metacentric height (GM_L for pitch and GM_t for roll). Since the inertia mass was not given, the equation

was further simplified using the radius of gyration in pitch or roll (k_{yy} or k_{xx}) which is equivalent to Formula 2.2.

$$k = \sqrt{\frac{I}{m}} \quad (2.2)$$

$$T = 2\pi\sqrt{\frac{I}{mgGM}} \leftrightarrow T = 2\pi\sqrt{\frac{k_{yy}^2}{gGM_L}} \quad (2.3)$$

For pitch, $k = k_{yy}$ and $GM = GM_L$, whereas for roll, $k = k_{xx}$ and $GM = GM_T$.

This formula is used with results given by the hydrostatic simulations which were provided by Técnico Solar Boat (TSB), where GM_L is 21.515 m and GM_t is 1.867 m. Also, the radius of gyration is calculated based on the longitude (L=8 m) and width/beam (B=2.5 m). Based on the International Towing Tank Conference (ITTC) recommendation [16], when the radius of gyration for pitch and roll is unknown a value of 0.25L and 0.35B to 0.40B, respectively, should be used.

Based on the given values, the pitch and roll natural period is 0.8654 and 1.2853 respectively, which is equivalent to 1.1555 Hz and 0.7780 Hz. These values represent a estimative of the natural frequencies of the vessel.

As for the frequency of the waves it has an extensive range and depends on several factors. Studies on this subject show higher energy for waves with 1 to 30 seconds periods which is equivalent to 0.033 Hz to 1 Hz, Figure 2.5.

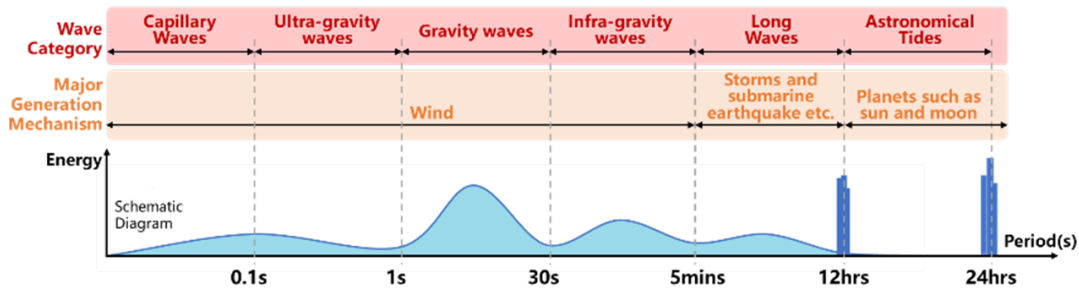


Figure 2.5: Classification of waves by period and respective energy [1].

The environmental excitation experienced by SG01 was further investigated using experimental data collected while the vessel was adrift at sea. Ten datasets containing the roll and pitch angles were collected and combined to obtain the frequency domain response shown in Figure 2.6. The results show significant spectral components at approximately 0.05 Hz and 0.5 Hz. The lower frequency must be due to the fast winds or transient oscillations, and it is consistent with the local forecast wave period. The higher frequency can be caused by the natural frequency of vessel. The root causes were not investigated further, since it falls outside the scope of this project.

Both theoretical and experimental results predict or measure frequency up to 1 Hz. In the context

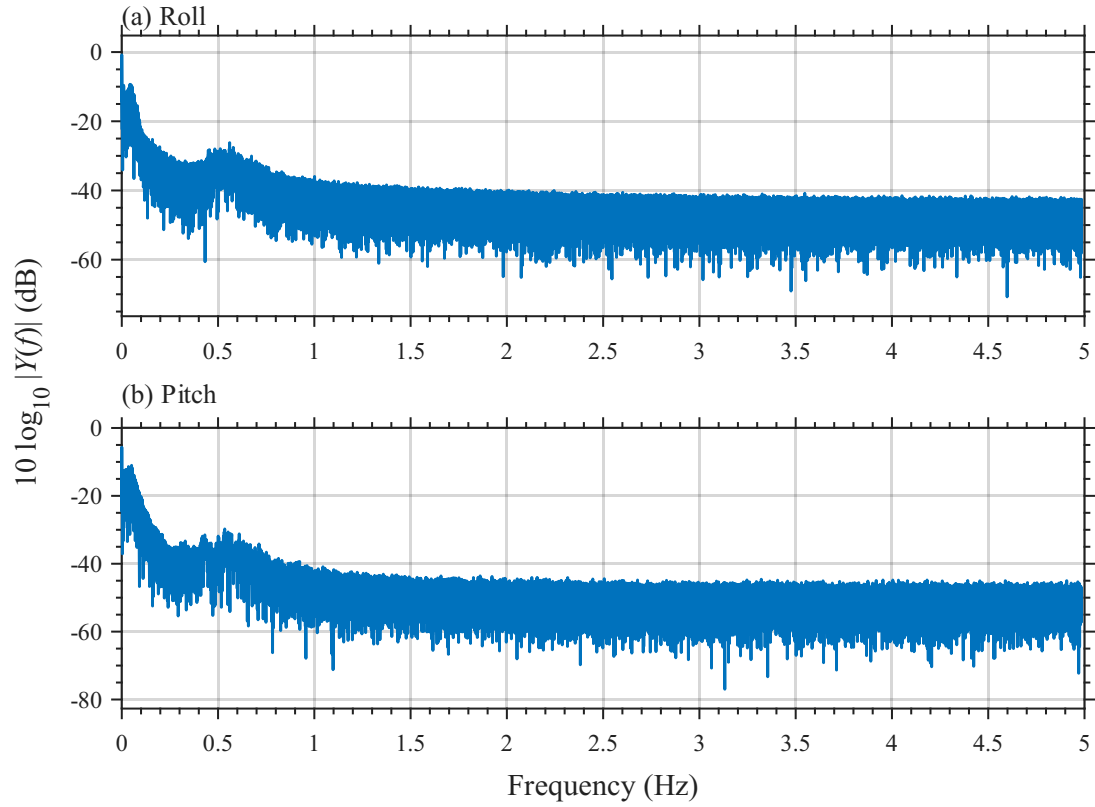


Figure 2.6: Measured oscillations from SG01.

of the MPPT algorithm, the goal will be a minimum speed of 1 Hz or 1 seconds and a typical speed of more than 2 Hz or 0.5 seconds.

These value will have directly influence on the sampling time as will discuss later.

3

State-of-the-Art

Contents

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3.4	Types of DC-DC Converters	25
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In this chapter, the main State-of-the-Art concepts regarding the engineering behind an MPPT charger and solar panels are explained. More specifically, it addresses key challenges such as: what is a solar panel, MPPT algorithms, DC-DC converter types, and battery charging techniques. Additionally, it contains a market analysis.

3.1 Solar Panels

Photovoltaic (PV) panels are devices that convert solar irradiance into electrical energy. A PV panel consists of multiple solar cells electrically connected in series and/or parallel to achieve the desired voltage and current levels. As in conventional electrical systems, the interconnection topology directly affects the electrical characteristics of the panel: series connections increase the output voltage, while parallel connections increase the output current [17].

PV panels connected in series operate at higher voltage levels, which reduces conduction losses in interconnecting cables and improves the efficiency of the DC-DC conversion stage. Moreover, power converters designed for high voltage and low current operation are generally less complex and more cost-effective than those operating at low voltage and high current. However, series configurations are more sensitive to partial shading or panel failure, as the output current of the entire string is limited by the weakest module [18]. In contrast, parallel configurations provide improved tolerance to shading and faults but result in higher current levels and increased conduction losses. In this project, series/parallel configuration of the solar panels will not be changed, since it is already pre-defined by Técnico Solar Boat (TSB), but it is important to understand the implications of each configuration.

Each cell is made of a semiconductor material, usually silicon. This semiconductor is doped with phosphorus, a group V element, to create a negative type layer. On the other side, a layer is doped with boron, a group III element, to create a positive type layer. This creates a p-n junction, which is essential for the photovoltaic effect [3].

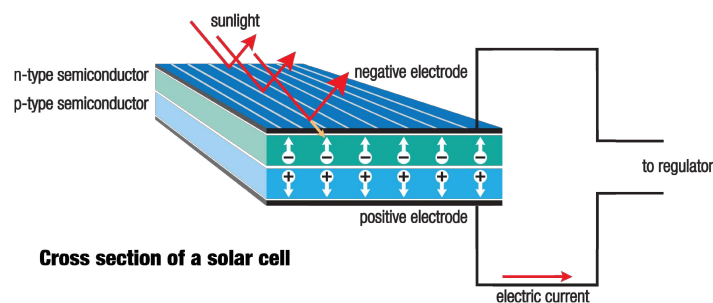


Figure 3.1: Principle of operation of a solar panel [2].

When sunlight arrives at the solar cell, the photons energy is absorbed by the semiconductor material. This energy excites electrons, allowing them to escape their atomic bonds and create electron hole

pairs. The electric field at the p-n junction drives these free electrons towards the n-type layer and holes towards the p-type layer, generating a flow of electric current when the cell is connected to an external circuit [3], Figure 3.1.

The energy generated results in voltage and current outputs that are not constant. Voltage and current are interdependent, therefore, a variation in one parameter leads to a corresponding variation in the other. This relationship is nonlinear and can be characterized by a current-voltage (I-V) curve. Furthermore, the power output also varies with operating conditions and can be represented by a power-voltage (P-V) curve. Both curves are represented in Figure 3.2. In both curves, there is a specific point where the power output is maximized, known as the Maximum Power Point (MPP). This point corresponds to a unique combination of voltage and current, denoted as V_{mpp} and I_{mpp} , respectively. Operating the solar panel at this point ensures optimal energy extraction under a given condition.

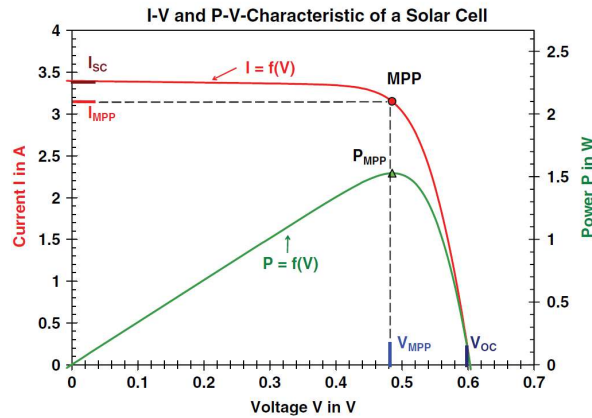
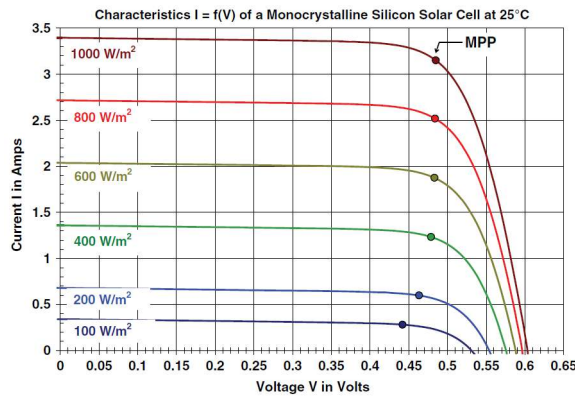


Figure 3.2: I-V and P-V curves of a solar panel [3].

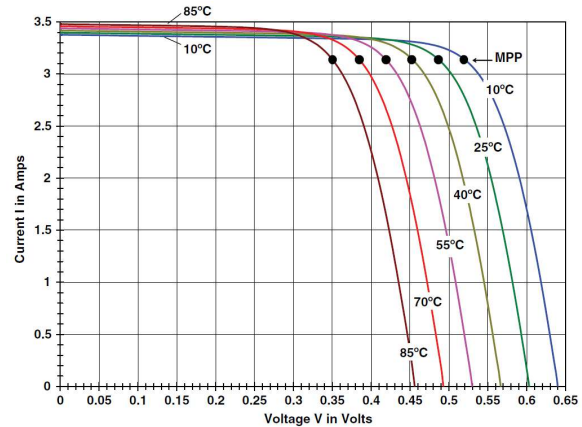
The characteristic curves of a solar panel depend on various factors, including the quality of the materials used, the solar cells, and environmental conditions such as temperature, solar irradiance, wind, and shades. It is noted that under partial shading conditions, it is possible to have multiple local maximums, but overall, there is still only one true MPP [19].

As seen in Figure 3.3a, the variation of the solar irradiance changes the I-V and P-V curves of the solar panel. The increase in irradiance produces more power by mainly increasing the current output of the panel. On the other hand, the temperature has an opposite effect (Figure 3.3b). The increase in temperature produces a decrease in power output, mainly by decreasing the voltage output of the panel.

Both environmental conditions have a meaningful impact on the MPP of the solar panel and therefore on the power output of the system. To maximize the power extracted by the solar panels, a classic power converter, like a buck converter with fixed duty cycle, will not be efficient since neither the current nor the voltage are constant for different conditions. Instead, an MPPT charger is used, which will find the MPP and operate the solar panel at that point while sensing the changes caused by the environment.



(a) Cell temperature of 25 °C and variable Irradiance.



(b) Irradiance of 1 kW/m² and variable temperature.

Figure 3.3: I-V curves under different conditions [3].

3.2 Maximum Power Point Tracker (MPPT)

There are two main types of "trackers", the mechanical trackers and the electrical trackers. Both aim to maximize the energy extracted from a photovoltaic panel, but in different ways. Mechanical trackers are movable structures that follow the sun's movement, maximizing the panel's exposure to sunlight throughout the day. These devices are complex, expensive, and inefficient, which is why most research has focused on electrical "tracking" of the MPP.

An electrical Maximum Power Point Tracker (MPPT) charger is a device that is used to optimize the power output from solar panels by continuously tracking and adjusting the operation point of the panels to ensure they operate at their MPP.

To achieve its goal, an MPPT charger is usually composed of a DC-DC converter, a microcontroller, and some sensors (Figure 3.4). The sensors are used to measure the input/output variables of the control system (typically voltage and current). The information acquired by these sensors is processed by the microcontroller, which runs an algorithm to determine if the MPP was reached or how to act on the system towards the MPP. In the second case, the microcontroller generates a signal to control the DC-DC converter that will adjust its operation accordingly.

The output can be connected to different types of loads, like batteries, DC loads or inverters. This project focuses on batter charging applications.

3.3 MPPT Algorithms

There are plenty of MPPT algorithms, each one with its own advantages and disadvantages. The choice of the algorithm will depend on the specific application, the desired performance, and the available

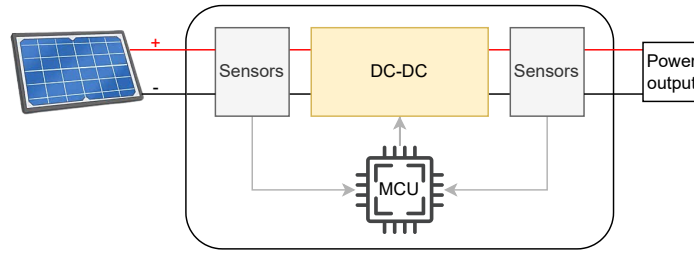


Figure 3.4: Simplified structure of an MPPT charger.

resources. In this work, the aim is to extract the maximum energy possible of a solar panel installed in a moving boat, which will produce an I-V curve variation due to the quick changes in irradiance caused by changes of inclination, orientation and clouds, and also temperature changes due to waves and wind. For these reasons, tracking speed and accuracy are the most important factors to consider.

With this in mind, the most relevant algorithms are briefly reviewed in order to identify the most suitable option for this project.

3.3.1 Constant Voltage (CV)

This is the simplest and most inefficient method. It works by operating the solar panel at a fixed voltage. This referenced voltage is used to calculate the duty cycle of the DC-DC converter that will maintain the operating point of the solar panel.

Well known variants improve performance by setting the reference voltage to a fraction of the open-circuit voltage (typically 71-78%) [19]. This voltage is obtained by briefly disconnecting the panel from the load, increasing reference accuracy and efficiency. Another variant uses a fraction of the short-circuit current as the reference [20].

Although both methods are simple and cost-effective, they are inherently inefficient due to the necessity of interrupting energy production during measurement phases. However, alternative implementations address this limitation through proxy measurement techniques employing dummy cells or diodes, whose physical properties closely approximate those of standard solar cells, thereby enabling continuous power generation during measurement acquisition [19] [21]. Still, the MPP is not always located at the same percentage of the open circuit voltage or short circuit current, leading to a lack of accuracy and therefore lower efficiency [14].

3.3.2 Perturb and Observe (P&O)

The Perturb and Observe (P&O) method is widely used in commercial products and is the basis of many advanced algorithms. Its popularity lies in its simplicity, low cost, and ease of implementation.

As its name indicates, the algorithm perturbs the voltage of a PV array and observes the resulting effect on the output power [22]. If an increase in voltage leads to an increase in output power, the operating point is moving toward the MPP, and the algorithm continues to increase the voltage. Conversely, if the output power decreases, the operating point is moving away from the MPP, and the direction of the perturbation is reversed [4, 23]. Figure 3.5 illustrates the flowchart of the algorithm.

The biggest drawback of this algorithm is that it oscillates around the MPP, which produces power losses. This oscillation can be reduced by decreasing the step size of the perturbation, but this will also reduce the tracking speed of the algorithm. This results in an inherent trade-off between tracking speed and accuracy [19, 20].

Another well known issue of P&O is that it can get confused in rapidly changing environmental conditions, like fast irradiance changes caused by clouds. In this case, the algorithm can misinterpret the power change caused by the environmental variation as a result of its own perturbation, leading it to move away from the MPP instead of towards it [19]. For example, if the perturbation was in the wrong direction, but the irradiance increased, the power output would increase, and the algorithm would continue perturbing in the wrong direction (Figure 3.6).

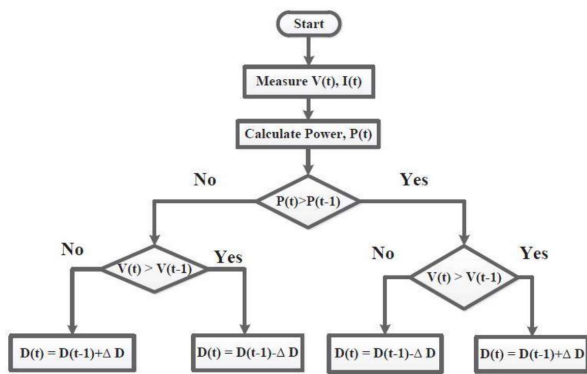


Figure 3.5: Flow chart of the classic version of Perturb and Observe algorithm [4].

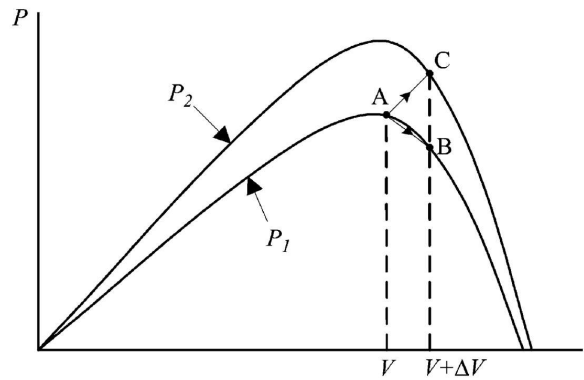


Figure 3.6: Divergence of P&O from MPP as shown in [5].

To address this limitation, an additional condition can be incorporated into the algorithm that evaluates two consecutive measurements of ΔP and ΔV . Specifically, when the signs of consecutive ΔP measurements do not follow the signs of ΔV , this indicates that the observed power variation results from environmental disturbances rather than the algorithm's perturbation. Consequently, the voltage adjustment is suppressed. This refined variant is designated the two-point algorithm or improved P&O method [17].

There are other variations of P&O algorithm that improve its performance, like Variable Step Size P&O (VSS-P&O), the three-point P&O, a-factor P&O, and more. Unfortunately, due to the scope of this work, it is not possible to explain all of them.

3.3.3 Incremental Conductance (IncCond)

The Incremental Conductance (IncCond) algorithm is another widely used MPPT method due to its accuracy and ability to track the MPP under rapidly changing environmental conditions.

This algorithm is based on the fact that at the MPP, the derivative of power with respect to voltage is zero. By knowing the output voltage and current of the solar panel, the algorithm can calculate the conductance (I/V) and the incremental conductance (dI/dV) [24].

The following formulas, Equation 3.1, show the equation on which this algorithm is based.

$$\frac{dP}{dV} = \frac{d(V \cdot I)}{dV} = I + V \frac{dI}{dV} \rightarrow \frac{1}{V} \times \frac{dP}{dV} = \frac{I}{V} + \frac{dI}{dV} \quad (3.1)$$

At the MPP point, the slope of the P-V curve is zero, which means that the negative of the conductance, $-I/V$, is equal to the incremental conductance, dI/dV , (Figure 3.8). The algorithm compares these two values to determine if the operating point is at, to the left, or to the right of the MPP [20] [19]:

$$\begin{aligned} \frac{dP}{dV} = 0 & \iff \frac{dI}{dV} = -\frac{I}{V} & \text{at MPP} \\ \frac{dP}{dV} > 0 & \iff \frac{dI}{dV} > -\frac{I}{V} & \text{left of MPP} \\ \frac{dP}{dV} < 0 & \iff \frac{dI}{dV} < -\frac{I}{V} & \text{right of MPP} \end{aligned} \quad (3.2)$$

Based on these conditions, the algorithm adjust its operating point (increasing or decreasing the voltage) using a variable step size based on the instantaneous conductance relative to the incremental conductance, Figure 3.7. For example, as suggested in [19], the step size can be calculated using a proportional integral (PI) controller with zero as input reference and error defined as:

$$e = \frac{I}{V} + \frac{dI}{dV} \quad (3.3)$$

Comparative analysis demonstrates that the IncCond algorithm exhibits superior performance relative to the P&O method. This outcome is expected, as the Incremental Conductance technique was developed specifically to address the limitations inherent in the P&O approach [24]. For example, the IncCond method eliminates the oscillation problem around the MPP that is characteristic of the P&O method. Additionally, IncCond demonstrates superior performance under rapidly changing environmental conditions, such as sudden irradiance variations, since it uses the derivative information that is less sensitive to transient disturbances [17]. The downside is the increase complexity and therefore the number of parameters increase too which means that the simulation time will increase too. Also, it uses fraction which can be very demanding for microcontrollers.

Note: Figure 3.7 was made for a buck converter, so the duty cycle actions are inverted for a boost converter. For example, if the operating point is to the left of the MPP, the duty cycle should be decreased

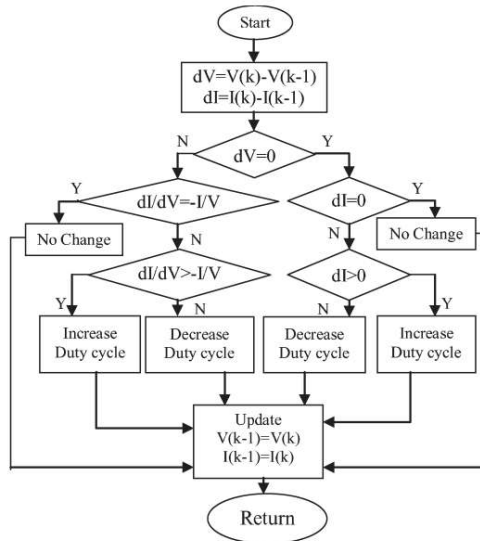


Figure 3.7: Flowchart of the IncCond method with direct control [6].

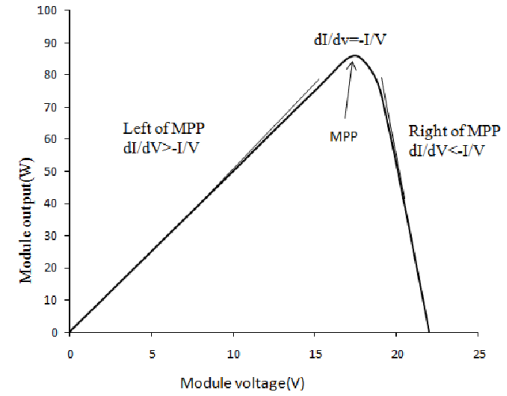


Figure 3.8: IncCond method principle.

to increase the voltage of the solar panel and move towards the MPP.

3.3.4 Look Up Table (LUT)

The Look-Up Table (LUT) method is a simple and fast MPPT algorithm that relies on pre-calculated/measured data to determine the optimal operating point of a solar panel [20].

The LUT table contains several entries organized by voltage and current. Each of these entries contains the optimal duty cycle of the DC-DC converter, that are pre-calculated or measured for a specific voltage and current. This approach allows the algorithm to reach the MPP in one clock cycle by measuring the voltage and current of the solar panel, looking for the closest entry in the LUT table, and applying the corresponding duty cycle to the converter. The entries of the table can be calculated through a mathematical model of the solar panel with different conditions (temperature and irradiance) or measured in a laboratory.

Table 3.1 is a visual representation of a 2D LUT, only based on current and voltage. Each entry represents the duty cycle used in each case.

The main advantage of this method is its speed since it can reach the MPP in one clock cycle. This makes it suitable for applications where fast tracking is required, or weak microcontrollers are in use. However, it is not very accurate [4].

To improve the accuracy, the LUT table needs to be larger, which increases the required memory resources. Also, by adding the temperature and/or irradiance to the table would increase the accuracy, making it almost 100% accurate, but this would increase the size of the table exponentially. In extreme

Table 3.1: LUT matrix representation with $m \times n$ elements.

	V_1	$\dots$	V_k	$\dots$	V_m
I_1	D_{11}	$\dots$	D_{1k}	$\dots$	D_{1m}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_j	D_{j1}	$\dots$	D_{jk}	$\dots$	D_{jm}
$\vdots$	$\vdots$		$\vdots$		$\vdots$
I_n	D_{n1}	$\dots$	D_{nk}	$\dots$	D_{nm}

cases, the memory requirements became expensive and slow. There is a trade-off between size, speed, and accuracy that needs to be considered when using this method. Also, the LUT method does not adapt to the aging of the solar panel or changes in its characteristics over time, unless the table is updated periodically through new measurements, which adds maintenance time.

3.4 Types of DC-DC Converters

The DC-DC converter is a crucial component of an MPPT charger, as it allows the adjustment of the operating point of the solar panel to match the MPP. His choice is highly important to the project since it directly affects the efficiency, cost, and complexity of the overall system.

DC-DC converters can be classified into two main categories: isolated and non-isolated converters. The main difference between these two types of converters is the presence or absence of galvanic isolation between the input and output circuits [25].

3.4.1 Non-isolated versus isolated DC-DC converters

Th3 galvanic isolation is usually achieved by using a transformer, which provides electrical isolation between the input and output sides of the converter. This isolation is important in applications where safety is a concern, such as in medical devices or industrial equipment, as it helps to prevent electrical shock and damage to sensitive components. In the context of solar energy systems and E-mobility, isolated converters are often used when the solar panel is connected to the grid [26], as they help to protect against voltage spikes and other electrical disturbances. Also, non-isolated converter can provide very higher/lower gain, which is useful for applications where solar panels MPP voltage is very different from the battery or load voltage (typically medium/high voltages which is not the case).

In contrast, non-isolated converters do not provide galvanic isolation between the input and output circuits. However, they generally offer higher efficiency, lower cost, reduced size, and lower circuit complexity (Table 3.2). Additionally, they present fewer challenges in thermal management and are

simpler to design. These advantages, together with the fact that galvanic isolation is not a requirement for this application, led to the exclusion of isolated DC-DC converters from this project.

Table 3.2: Comparison between Non-isolated and isolated converters.

Type	Non-isolated Converters	Galvanic Isolated Converters
Circuit Complexity	Low	High
Efficiency	High	Medium
Size	Compact	Large
Electromagnetic Interference (EMI)	Medium/High	High
Switching Frequency	Medium	Medium
Thermal Management	Easier	Moderate
Control Complexity	Moderate	Moderate
Gain	Near 1	Very High/low
Cost	Low	High

Legend: undesired medium good

3.4.2 Non-isolated DC-DC converters

Modern non-isolated conversion circuits generally use one of three basic topologies: buck, boost, or buck-boost converters. They are "basic" in the sense that only one switching element is needed. A given topology is used to obtain a specific result, such as voltage step-down, voltage step-up, or hybrid mode [8].

The **boost converter** is a step-up DC-DC converter widely employed in MPPT controllers when the required output voltage is higher than the solar panel open circuit voltage. It uses an inductor to store energy when the transistor is in the ON state and a diode in reverse biased (Figure 3.9b). When the transistor is on the OFF state, the inductor releases the stored energy to the output through the diode, increasing the output voltage [27].

This topology has been extensively documented in the literature and is distinguished by its inherent simplicity, cost-effectiveness, and superior efficiency characteristics [12] [4] [24]. Depending on the design variation used and constraints, it can achieve conversion efficiencies up to 98% [28].

A limitation of this topology is that it cannot emulate an impedance lower than the load impedance, which prevents it from reaching values close to the short-circuit current of the PV module. Consequently, it is not compatible with all solar panels or MPPT algorithms [28, 29].

The **buck converter** is a step-down DC-DC converter also widely used for applications where the solar panel voltage is higher than the battery or load voltage. Similar to the boost converter, it uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. However, in this case, the input is connected to the output when the transistor is in the ON state. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, decreasing the output voltage (Figure 3.9a) [28].

This converter also has a similar downside to the boost converter. The buck converter can not emulate a smaller impedance than the load impedance, and therefore, it can not reach values near the open circuit voltage of the PV module [28] [29].

Buck-Boost converters are step-up and step-down DC-DC converters that can operate in both modes. This makes them more versatile than the previous two topologies, but also more complex and less efficient [28]. This converter uses an inductor to store energy when the transistor is in the ON state and the diode is reverse-biased. When the transistor is in the OFF state, the inductor releases the stored energy to the output through the diode, increasing or decreasing the output voltage (Figure 3.9c) [8].

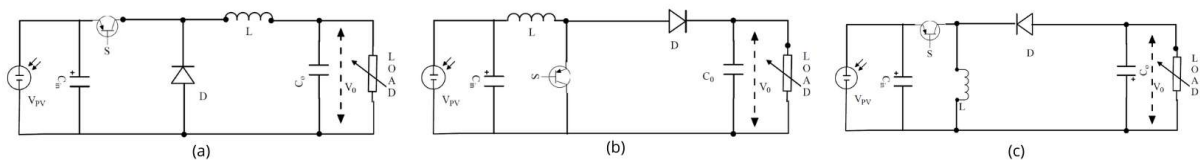


Figure 3.9: Buck (a), Boost (b) and Buck-Boost (c) DC-DC converters.

The conventional buck-boost converter produces an output voltage inverted with respect to the input which creates challenges on sensor measurements, driving circuit and isolation, consequently, it is unsuitable for the battery charging system [27].

An enhanced variant of the buck-boost converter is the CUK converter, illustrated in Figure 3.10a. The main characteristic of the CUK converter is that the currents through inductors L_1 and L_2 are equal, allowing the use of a common magnetic core, which helps reduce current ripple. However, a major drawback of this topology is that the entire load current must pass through the coupling capacitor C . In addition, the converter produces an inverted output voltage [27].

The issue of output voltage polarity inversion can be addressed by using a single-ended primary inductor converter (SEPIC), as shown in Figure 3.10b. Nevertheless, its main drawbacks are the high output current and voltage ripple which increases losses and electromagnetic interference (EMI) [30]. These limitations are mitigated by the ZETA converter, shown in Figure 3.10c, which provides improved output current characteristics while maintaining the same output voltage polarity as the SEPIC converter [23] [27].

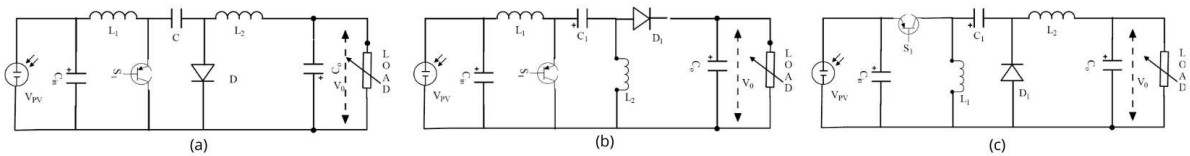


Figure 3.10: CUK (a), SEPIC (b) and ZETA (c) DC-DC converters.

3.4.3 Comparison between Non-isolated DC-DC converters

All the non-isolated DC-DC topologies explained before have their own advantages and disadvantages. The choice of the topology will depend on the specific application, the desired performance, and the available resources. To help with this choice, Table 3.3 shows a comparison between the most relevant characteristics of each topology [30]. The most significant advantages of each topology are highlighted accordingly with the aim of the project.

Table 3.3: Comparison of non-isolated DC-DC topologies.

Topology	Boost	Buck	Buck-Boost	ZETA	CUK	SEPIC
Electrical performance						
Efficiency	High	High	High	Medium/High	Medium	Medium
EMI	Low	Low	High	Medium	Medium	Medium
Input current ripple	Low	High	High	Low	Very low	Medium
Output current ripple	High	Low	High	Very low	Low	Medium
Output voltage ripple	Medium	Low	Medium	Low	Medium	High
Structural characteristics						
Circuit complexity	Low	Low	Low	Medium	Medium	Medium
Size	Low	Low	Low	Medium	Medium	Medium
Storage elements	1 L	1 L	1 L	2 L, 1 C	2 L, 1 C	2 L, 1 C
Switch stress	No	Yes	No	Yes	Yes	Yes
Functional aspects						
Type	Step-Up	Step-Down	Step-Down/Up	Step-Down/Up	Step-Down/Up	Step-Down/Up
Versatility	Low	Low	Medium	High	Medium	High
Inverted output	No	No	Yes	No	Yes	No
Reliability	High	High	High	Medium	Medium	Medium
Economic						
Cost	Low	Low	Low	Medium	Medium	Medium

Legend: good important

For MPPT controllers, the choice is mainly governed by the possibility of operating at the MPP which is connected with the converter type (Step Up, Down or Down/Up). Also, efficiency, cost (related to the number of storage elements), output voltage ripple are relevant in this choice.

The buck converter can be excluded from this comparison since in this project the solar panel voltage is lower than the battery voltage, therefore, a step-down converter is not suitable. However, when choosing between the boost and the step-up/down converters, there is a trade-off. The boost converter is simpler, cheaper and more efficient. On the other hand the step-down/up converters are more versatile since they can operate in a wide range of input voltages, which is interesting since the size of the solar panels and battery voltage can change.

Considering that in this project the solar panel voltage is always lower than the battery voltage, the boost converter is the most suitable topology due to its simplicity, cost-effectiveness, and high efficiency. However, if versatility were to become a priority, the ZETA converter would be the preferred choice among the step-up/down topologies, owing to its non-inverted output voltage and low output current ripple, which are advantageous for battery charging applications.

3.5 Battery charging techniques

As the demand for electronic devices and E-vehicles increases, the need for efficient, compact, and lightweight batteries has emerged. Among many existing technologies, lithium-ion batteries have one of the best energy-to-weight/volume ratios and, at this moment, it is the technology used in TSB batteries. But, like every other battery technology, they need to be charged properly to ensure their safety and longevity [7].

To charge these batteries properly, the MPPT algorithm can operate alone if an additional converter is used in series. But in most of MPPT chargers, the MPPT and the battery charging unit are integrated in the same converter. This way, the MPPT algorithm can adjust the operating point of the solar panel to extract the maximum power while the battery charging technique ensures that the battery is charged properly. A proper combination of both algorithms/techniques is essential to ensure both maximum energy extraction and safe battery charge.

There are several techniques to charge lithium-ion batteries, the most relevant for this use case being the constant current-constant voltage (CC-CV) method [31]. Also, it is widely used in commercial chargers due to its simplicity and effectiveness [32]. This method consists of two main phases (Figure 3.11). In the first phase, the battery is charged with a constant current until it reaches the maximum voltage of the battery. In the second phase, the voltage is held constant at this limit while the current gradually decreases as the battery approaches full charge. Once the current drops below a certain threshold, the charging process is terminated to prevent overcharging [32].

When employing the CC-CV charging technique with a MPPT controller, the initial Constant Current (CC) phase is regulated by the MPPT algorithm, which extracts the maximum available power from the solar panel limited by the maximum charging current. If the maximum charge current is reached, it will operate at CC. Once the Constant Voltage (CV) phase is reached, the MPPT function is disabled, and the power converter increases the solar panel voltage to progressively reduce the charging current while keeping the output voltage constant.

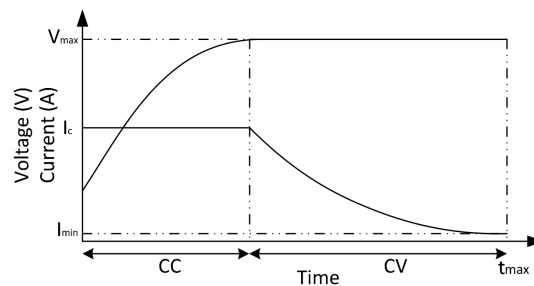


Figure 3.11: Charging profile of CC/CV [7].

More advanced high-performance battery charging techniques have been proposed in literature to reduce charging time without increasing the charging current, thereby preserving battery lifetime. However, these techniques generally introduce higher switching losses and increased system complexity. In the present application, the current generated by the photovoltaic panels is less than half of the maximum allowable battery charging current, consequently, the charging current is inherently limited by the solar panels rather than by the charging strategy. Furthermore, battery lifetime is not a critical concern in this project, as the battery is expected to undergo a limited number of charge cycles, not exceeding 20 cycles per year.

Therefore, the CC-CV charging method is considered the most suitable technique for this project, owing to its simplicity, effectiveness, and widespread adoption in commercial battery chargers.

Other charging techniques fall outside the scope of this work, however, some methods are briefly discussed in the references provided [31] [7] [32].

3.6 Commercial Options

To assess the feasibility of adopting a Off the shelf (OTS) solution, this section reviews existing commercial options for solar battery charging with MPPT capability. Both specialized Integrated Circuits (ICs) and complete commercial MPPT chargers are considered, and their electrical, functional, and economic characteristics are evaluated against the requirements of this project. The objective is to identify whether a commercially available solution can satisfy the constraints of the application or, alternatively, to justify the need for a custom designed converter and control system.

3.6.1 Specialized ICs

Several commercially available Integrated Circuit (IC) are specifically designed for solar battery charging and MPPT applications. These devices are relevant to this project since, if suitable, they could significantly reduce design complexity by integrating functions such as DC-DC conversion, protection mechanisms, and MPPT algorithm.

Table 3.4 summarizes four commonly used commercial ICs. Characteristics marked in red indicate incompatibilities with the project requirements, leading to the exclusion of the corresponding devices.

The BQ24650RVAT is a Stand-Alone Synchronous Buck Battery Charge Controller which implements a high-efficiency battery charger while also providing safety features. By adding a Microcontroller Unit (MCU) to run a MPPT algorithm and current and voltage sensors, this IC can track the MPP of a solar panel, making it appropriate for applications with 12/24 V batteries and high-power panels, such as electric vehicles. However, it does not support the required 48 V battery voltage and is therefore unsuitable for this work.

The MAX20801TPBA+ and SPV1040TTR target low-power IoT applications with small batteries and solar panels. Both devices are unable to meet the 48 V output requirement, and their limited publicly available information on tracking speed further restricts performance evaluation.

The LT8491IUKJ#PBF offers a wide input and output voltage range, buck-boost operation, high efficiency, communication capabilities, and CC-CV charging. Despite these advantages, its reported MPPT tracking speed of up to 2 s is insufficient for the dynamic conditions of this application. Additionally, its I²C interface would require an additional microcontroller to provide CAN communication, increasing system cost.

Table 3.4: Comparison of commercial specialized ICs.

Model	BQ24650RVAT	MAX20801TPBA+	SPV1040TTR	LT8491IUKJ#PBF
Electrical specifications				
Input voltage	5 to 28 V	1.5 to 18 V	0.3 to 5.5 V	6 to 80 V
Output voltage	12 to 24 V	12.4 V	-0.3 to 5.5 V	1.3 to 80 V
Max output current	Externally limited	12 A	1.8 A	10 A
Switching frequency	600 kHz	Unknown	100 kHz	100 to 400 kHz
Electrical efficiency	95%	99.1% (max)	80 to 95%	95 to 99%
MPPT and control				
Algorithm	User input	Unknown	Unknown	P&O
Tracking accuracy	User dependent	99.9% (max)	Unknown	Unknown
Tracking speed	User dependent	Unknown	Unknown	1 to 2 s
Functional aspects				
Topology	Synchronous Buck	Synchronous Buck	Synchronous Boost	Buck-Boost
Communication	No	No	No	I ² C
Configurability	No	No	No	Yes
Charge profile	CC-CV	Unknown	Unknown	CC-CV
Extra hardware needed				
Transistors	Yes	No	No	Yes
Sensors	Yes	No	No	No
Storage elements	Yes	Yes	Yes	Yes
Economic				
Price	5.25 €	4.50 €	3.02 €	15-20 €

Legend: unsuitable

3.6.2 Commercial MPPT chargers

Several commercial MPPT chargers are available that can address the problem under analysis without requiring additional hardware. However, each solution presents limitations when evaluated against the project requirements. Table 3.5 summarizes selected commercial MPPT chargers, with red-marked characteristics indicating incompatibilities.

The GVB-8-Li-CV(50.4) is a high-performance and fast MPPT charger, particularly suited for dynamic environments such as marine applications. It has been successfully used by TSB in previous projects, demonstrating high efficiency and reliability. Nevertheless, its lack of configurability, fixed output voltage, no communication and high cost restrict its suitability for this application. Despite these drawbacks, it serves as a valuable reference design.

The Smart Solar MPPT 100/20, developed by Victron Energy, offers high efficiency, fast tracking, and integrated communication interfaces. However, it uses a buck topology that requires higher input voltages, making it incompatible with the small and distributed solar panel arrays used in the TSB application.

Other commercial solutions, such as the Rover Lite and MS4840N, provide acceptable performance but rely on communication protocols that are not compatible with the project requirements and offer limited technical documentation.

The Reboost V0.2.1 is an open source MPPT charger originally developed for automotive applications. While it provides flexibility and near adequate specifications, its reliance on expensive and discontinued components, along with the use of a boost topology, limits scalability and adaptability to potential future battery voltage changes.

Overall, none of the evaluated commercial MPPT chargers fully satisfy the technical, economic, and configurability requirements of this project, motivating the development of a custom solution.

Table 3.5: Comparison of commercial MPPT controllers.

Model	GVB-8-LI-CV (50.4V)	Smart Solar MPPT 100/20	Reboost V0.2.1	Rover Lite	MS4840N
General information					
Brand	Gensun	Victron Energy	TPEE	Renogy	BougeRV
Type	Boost	Buck	Synchronous Boost	Boost	Boost
Control and communication					
Communication	No	VE.can and Bluetooth	Yes	Bluetooth and RS485	Bluetooth
Configurable	No	Yes	Yes	Yes	Yes
GUI	No	Yes	Yes	Yes	Yes
MPPT and charging					
Prog. battery voltage	No	Yes	Yes	non-lithium batteries	Yes
Tracking speed	15 Hz / 66.6(6) ms	Fast (unknown)	Programmable	Unknown	Unknown
Tracking efficiency	99%+ typical	Unknown	Programmable	99%	Unknown
Electrical efficiency	96-99% typical	98% peak	Unknown	97%	Unknown
Charger profile	CC-CV	Unknown	Programmable	Unknown	CC-CV
Hardware implementation					
MCU	ATtiny461A-U	Unknown	STM32G474	Unknown	Unknown
MCU Details	8-bit, RISC, 64 MHz	—	32-bit, Arm, 170 MHz	—	—
Transistor	FZT951	Unknown	GS61008T	Unknown	Unknown
Transistor type	PNP BJT	Unknown	GaN FETs	Unknown	Unknown
Economic					
Price	240 €	100-200 €	Components: 250-280 € PCB: 20-100 € Total: 270-380 €	300-350 €	120-160 €

Legend: unsuitable

4

Implementation

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This chapter presents the implementation of the proposed MPPT charger. The charger is based on a synchronous boost converter controlled by an STM32 microcontroller, with dedicated sensing, protection, communication and configuration features.

The chapter first describes the DC–DC converter, including its operating principles and the dimensioning of the main power components. The processing unit and its configuration are then presented, followed by the voltage, current, temperature and irradiance sensing circuits. Finally, the firmware architecture is described, covering the charging state machine, communication interfaces and the graphical user interface used to configure the device.

4.1 DC-DC converter

Since the solar panels voltage of São Gabriel 01 (SG01) is always lower than the battery voltage, a step-up converter is required. This section presents its operating principles and the dimensioning of the main components.

4.1.1 Operation principles

The DC-DC converter chosen for this project was a boost converter. This topology uses one switching device, one inductor, one output capacitor and a diode, Figure 4.1. The switching device is used to control the output voltage by switching On and Off at a specified frequency (f_{sw}) and duty cycle (D). The inductor is used to store energy when the switch is on (stage 1) and release it when the switch is off (stage 2), while the output capacitor is used to smooth the output voltage, Figure 4.1.

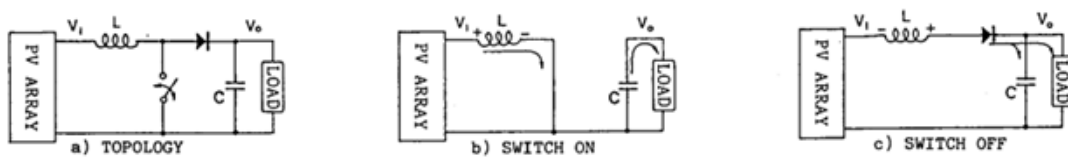


Figure 4.1: Classic Boost converter topology and stages [8].

Analyzing the boost converter separately on his two stages, it is possible to define the drop voltage across the inductor (V_L), as a function of the input voltage (V_{in}) and output voltage (V_{out}) as follows:

$$V_L = \begin{cases} V_{in} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} & \text{if } t \in [D \cdot T, T] \end{cases} \quad (4.1)$$

With equation 4.1 and the fact that at stationary state the average voltage across the inductor is zero, it is possible to define the output voltage in function of the input voltage and duty cycle as follows:

$$V_{Lavg} = \frac{1}{T} \left(\int_0^{DT} V_{in} dt + \int_{DT}^T (V_{in} - V_{out}) dt \right) = 0 \Rightarrow V_{out} = \frac{V_{in}}{1-D} \quad (4.2)$$

Equation 4.2 is one of the most important equations in boost converter design, as it defines the relationship between the input voltage, output voltage and duty cycle. As the duty cycle increases, the output voltage also increases, which is the main principle behind boost converters. For our use case, it is import to analyze the relation between the input and output current as well. Considering no power losses, it is possible to define the input current as follows:

$$P_{in} = P_{out} \Leftrightarrow V_{in} \cdot I_{in} = V_{out} \cdot I_{out} \Leftrightarrow I_{in} = \frac{V_{out}}{V_{in}} \cdot I_{out} \Leftrightarrow I_{in} = \frac{I_{out}}{1-D} \quad (4.3)$$

However, these equations are only valid for ideal components since it does not take into account the parasitic elements of the components, such as the resistance of the inductor, the drain source resistance of the transistor, and the voltage drop across the diode (there are more parasitic components that affect the performance). In a real boost converter, these parasitic elements will cause a voltage drop and therefore reduce the output voltage.

Let's quickly analyze the influence of the inductor resistance (R_L) and the voltage drop across the diode (V_d) by changing the equation 4.1 to:

$$V_L = \begin{cases} V_{in} - R_{DS(on)} \cdot I_{DS} & \text{if } t \in [0, D \cdot T] \\ V_{in} - V_{out} - V_d & \text{if } t \in [D \cdot T, T] \end{cases} \quad (4.4)$$

In this case the at stationary state the average voltage across the inductor is $R_L \cdot I_L$ and therefore the output voltage can be defined as:

$$V_{Lavg} = \frac{1}{T} \left(\int_0^{DT} (V_{in} - R_{DS(on)} \cdot I_{DS}) dt + \int_{DT}^T (V_{in} - V_{out} - V_d) dt \right) \Leftrightarrow \quad (4.5)$$

$$\Leftrightarrow R_L \cdot I_L = \frac{1}{T} [T \cdot D \cdot (V_{in} - R_{DS(on)} \cdot I_{DS}) + T \cdot (1-D) \cdot (V_{in} - V_{out} - V_d)] \Leftrightarrow \quad (4.6)$$

$$\Leftrightarrow V_{out} = \frac{V_{in} - D \cdot R_{DS(on)} \cdot I_{DS} - R_L \cdot I_L}{1-D} - V_d \quad (4.7)$$

Equation 4.7 shows the 3 causes of losses in a boost converter. The main cause is the voltage drop across the diode, which causes a significant reduction in the output voltage (usually 0.3 V to 0.7 V). This will be solved by using a synchronous boost converter (Figure 4.2), which replaces the diode with a transistor. In synchronous boost converters the voltage drop across the transistor is much smaller due to the low $R_{DS,on}$, usually lower than 0.1 V.

As for the inductor internal resistance, it is not possible to remove it entirely, so it must be taken into

account when designing the boost converter. The same is true for the drain source resistance of the transistor, which is also removable. Therefore, it is important to select components with low internal resistance to minimize the losses and increase the efficiency of the converter.

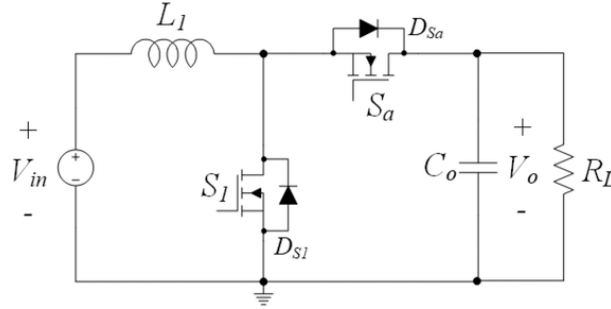


Figure 4.2: Synchronous boost converter topology [9].

4.1.2 Dimensioning

Let us first define the design goals of the synchronous boost converter. The converter must be able to step up the voltage from the solar panels, which may consist of a maximum of 40 cells (Maxeon Gen 3 from SunPower) connected in series, a minimum of 20 cells in series, and is optimized for 35 cells (Figure 4.1). The converter must charge a lithium-ion battery with a nominal voltage of 43.2 V. The switching frequency was defined as 100 kHz. This relatively high frequency allows the use of smaller inductors and capacitors, while still avoiding excessive switching losses.

Table 4.2 and 4.1 summarizes the system requirements.

Table 4.1: PV panel parameters at MPPT.

Configuration	V_{mppt} (V)	I_{mppt} (A)	P_{mppt} (W)	V_{oc} (V)	I_{sc} (A)
40 cells	24.68	6.05	139.6	29	6.374
35 cells	21.875	6.05	122.15	25.375	6.374
20 cells	12.34	6.05	69.8	14.5	6.374

Table 4.2: Converter design constraints.

Parameter	Value
Battery voltage (nominal)	43.2 V
Battery voltage (max)	50.4 V
Battery voltage (min)	30 V
ΔV_o	0.1 V
ΔV_i	0.5 V
Efficiency	> 90%

Since the load is a battery, the output ripple voltage must be limited to avoid aging and heating. Also, the voltage output voltage ripple must be low to avoid unexpected behavior of the electronic components

that are also connected to the battery. As for the input (the solar panel) the voltage ripple must be low for easier measurement and precise control. Therefore, the values presented in Table 4.2 were selected.

4.1.2.A Inductor

The inductor is one of the most important components in the boost converter since it is responsible for storing energy and releasing it to the load.

The choice of an inductor will highly influence the performance of the converter. High inductance will generate low current ripple, however it generates losses due to its internal resistance. On the other hand a smaller inductance will generate high current ripples, high stress in components and high magnetic interference. These current ripples can cause the system to operate in Discontinuous Conduction Mode (DCM). In DCM the inductor current reaches zero during the switching period, which means that the energy stored in the inductor is not enough to supply the load during the off time. This can cause high current and voltage ripples, which can damage the components and reduce the efficiency of the converter. Therefore, it is important to ensure that the inductance value is high enough to operate in Continuous Conduction Mode (CCM).

The equation that describes the inductor is defined as:

$$V_L = L \cdot \frac{di_L}{dt} = L \cdot \frac{\Delta I_L}{\Delta t} \Leftrightarrow L = \frac{V_L \Delta t}{\Delta I_L} \quad (4.8)$$

In the first stage (transistor in conduction mode), the voltage in the inductor is equal to the input voltage and it occurs for $D \cdot T$ seconds. Therefore, the inductor ripple current can be defined as follows:

$$\Delta I_L = \frac{V_{in} \cdot D \cdot T}{L} \quad (4.9)$$

As for the current ripple, it can be described using the ripple factor and the average current in the inductor as follows.

$$r = \frac{\Delta I_L}{I_{L,avg}} \quad (4.10)$$

$$I_{L,avg} = \frac{I_o}{1-D} \quad (4.11)$$

Using Equation 4.11, 4.10, 4.8 and 4.2, the following equation is achieved:

$$L = \frac{V_L \Delta t}{\Delta I_L} = \frac{V_{in} D T}{\frac{I_o r}{1-D}} = \frac{V_{in} (1-D) D T}{I_o \cdot r} = \frac{V_{out} (1-D)^2 D T}{I_o \cdot r} \quad (4.12)$$

The value of D that maximizes Equation 4.12 is 1/3, which results in the critical inductance formula:

$$MAX[(1 - D)^2 \cdot D] = (1 - 1/3)^2 \cdot 1/3 = \frac{4}{27} \quad (4.13)$$

$$L_{critical} = \frac{4}{27} \cdot \frac{V_o}{I_o \cdot r \cdot f} \quad (4.14)$$

Using the maximum battery voltage (50.4 V) and the maximum current ($I_{bat} + Margin = 10 \text{ A}$), and a ripple factor of 2 which is equivalent to the CCM limit, the following result is achieved:

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 2 \cdot 100000} = 3.73 \mu H \quad (4.15)$$

This is a reference value, if used, it would highly stress the components, generating heat, magnetic interference and high current ripples. Therefore, the calculation was repeated, this time with a ripple factor of 0.4, which is more reasonable.

$$L_{critical} = \frac{4}{27} \cdot \frac{50.4}{10 \cdot 0.4 \cdot 100000} = 18.66 \mu H \quad (4.16)$$

As it is explained in the simulation section (Section 5), an higher value will be beneficial to reduce current ripple and noise. For that reason the final inductor value was defines as $68 \mu H$, which results in a ripple factor of 0.11.

To select the appropriate inductor, the rated current and the saturation current must be taken into account. The rated current is the maximum current that the inductor can handle without overheating. In this case the rated current must be higher that the maximum short circuit current of the biggest solar panel configuration (40 cells in series), which is 6.374 A. Some margin must be added to this value, so the rated current must be higher than 10 A.

The saturation current is the maximum current that the inductor can handle without saturating. When the inductor saturates, the inductance drops significantly, which can cause high current ripples and damage the components. Considering the Open circuit voltage of a 40 cells solar panel (worst case), the inductor ripple current is 4.26 A at 100% duty cycle. Therefore, the saturation current must be:

$$I_{sat} > I_{in} + \frac{\Delta I_L}{2} = 6.374 + \frac{4.26}{2} = 8.50 \text{ A} \quad (4.17)$$

It is physical impossible to have a solar panel on short circuit current and open circuit voltage. And for this reason the calculation is overdimensioned it is used as a reference instead of a strict limit. The inductor chosen was the 74437529203680 from Würth Elektronik, which has an inductance of $68 \mu H$, a rated current of 16 A and a saturation current of 7.2 A ($|\Delta L/L| < 10\%$) and 10.6 A ($|\Delta L/L| < 30\%$). The saturation current is too close with the objective, however since the inductance is higher than what is needed, the inductance would still be acceptable even at -30% inductance.

Additionally, it has a low Equivalent Series Resistance (ESR) of only $10.3 \text{ m}\Omega$, high self resonant

frequency of 5.7 MHz and high maximum voltage of 250 V.

4.1.2.B Capacitor

The output capacitor is responsible for smoothing the output voltage and reducing the voltage ripple. The capacitor electric characteristic can be described by the following equation:

$$i_c = C \frac{dv_o}{dt} \quad (4.18)$$

Based on this equation, the capacitor value can be calculated based on the desired voltage ripple and the load current as follows:

$$i_c = C_o \frac{\Delta V_o}{\Delta t} = C_o \frac{\Delta V_o}{D \cdot T} \Leftrightarrow C_o = \frac{I_o D}{\Delta V_o \cdot f} = \frac{I_{in}(1-D)D}{\Delta V_o \cdot f} \quad (4.19)$$

Using the maximum input current of 10 A, a duty cycle of 0.5 (value that maximizes the formula), a voltage ripple of 0.1 V and a switching frequency of 100 kHz, the following value is achieved:

$$C_o = \frac{10 \cdot (1 - 0.5) \cdot 0.5}{0.1 \cdot 100000} = 250 \mu F \quad (4.20)$$

The same calculation is applied for the input capacitor, but in this case the voltage ripple is not as critical as in the output capacitor, since the input voltage is not directly connected to the load. Therefore, a voltage ripple of 0.5 V was used:

$$C_{in} = i_{in} \cdot \frac{\Delta t}{\Delta V_i} = \frac{i_{in} \cdot D}{\Delta V_{in} \cdot f} = \frac{10 \cdot 0.5}{0.5 \cdot 100000} = 100 \mu F \quad (4.21)$$

To choose the appropriate capacitors, the voltage rating, current ripple and ESR must be taken into account. The voltage rating must be higher than:

$$V_{rating} > MAX(V_{in}) + \Delta V_{in} = 29 + 0.5 = 29.5 V \quad (4.22)$$

Where 29 is the maximum voltage of a solar panel (40 cells in series) and 0.5 is the voltage ripple defined in Table 4.2.

The current ripple must higher than the RMS input current, which can be calculated based on the RMS formula for triangle wave as follows:

$$I_{ripple} > \frac{\Delta I / 2}{\sqrt{3}} = \frac{4.26 / 2}{\sqrt{3}} = 1.23 A \quad (4.23)$$

As for the ESR the value must be as low as possible to reduce the power losses.

After analyzing the market, the input capacitor selected was the 870575875006 which is a $56 \mu F$ (2 in parallel) with a voltage rating of 63 V and a ripple current of 2.4 A each (total of 4.8 A). The main reason for this choice was his low ESR ($22 m\Omega$) compared to other capacitors with similar values. Additionally, by using two capacitors in parallel, the ESR is reduced even further and the current ripple that they with stand is higher and the reliability of the system is increased.

The same process was repeated for the output capacitor, but in this case the voltage rating must be higher than the maximum battery voltage, which is 50.4 V. The current ripple must be higher than the RMS output current, which can be calculated as follows:

$$V_{rating} > MAX(V_{out}) + \Delta V_{out} = 50.4 + 0.1 = 50.5 V \quad (4.24)$$

$$I_{ripple} > \sqrt{\frac{I_{in}^2}{4} + \frac{\Delta I^2}{12} \cdot (1 - D)^2} = \sqrt{\frac{6.374^2}{4} + \frac{4.26^2}{12} \cdot (1 - 0)^2} = 3.42 A \quad (4.25)$$

The ripple current in the output capacitor is much higher than in the input capacitor, and it is not a common value for electrolytic capacitor, therefore instead of using a single capacitor, multiple capacitors in parallel will be used to achieve the desired current ripple rating.

After simulations and market analysis, the output capacitor value were increased due to the output current ripple dependency on the output voltage ripple. The final capacitor chosen was the 860040875005 with $100 \mu F$ (5 in parallel, so $5 \times 100 \mu F$) with a voltage rating of 100V and a ripple current of 0.8 A each (total of 4 A). Some other options were considered, and compared for optimization in section 5.1.2.

Additionally, the output and input needs to filtered to reduce the high frequency noise generated by the switching. This is achieved by adding a small capacitor in parallel with the main capacitors. The values use are $1 \mu F$ and $0.1 \mu F$. For this use case ceramic capacitors are better since their stacked multilayer construction provides extremely low ESR and Equivalent Series Inductance (ESL) which over all let them maintain low impedance up into the MHz range.

4.1.2.C Transistor and gate driver

The transistor is the switching device on a synchronous boost converter. The choice of the transistor will highly influence the behavior and performance of the converter.

When selecting a transistor for a switching converter, not only the maximum rating mater but the switching speed and internal resistance are also quite important.

There are many types of transistors (IGBT, MOSFET, GaN, etc.), however, at this frequency (100 kHz) and power level the choice is between MOSFETs and GaN transistors (Figure 4.3). The GaN (Gallium Nitride) transistor technologies has appeared in 1993 in lab experiments, however it has only reached the market more recently, first for radio frequency products in 2004 and only in 2009 for power electronics. The advantage of GaN transistors is their low internal resistance and very fast switching speed, which

can significantly reduce the switching losses and therefore increasing the efficiency of the converter. Additionally, they have a high energy density and lower $V_{gs(on)}$. However, they are more expensive and require a precise gate driver to operate correctly because their $\max(V_{gs})$ is usually low.

In applications where the efficiency is a key factor in the design like solar inverters or Maximum Power Point Tracker (MPPT) charger, it is better to use the component that has lower Internal (On) Resistance (R_{dsOn}) and supports higher switching frequency. In this case, GaN would be a better fit. MOSFET would take the second place since MOSFETs can handle higher switching frequencies as compared to IGBTs [10].

There are a lot of options in the market, however there was one in specifically that stand out due to their compact size, while having integrated gate driver and two transistors in the same package. The Integrated Circuit (IC) is the EPC23102, which was designed for power applications, with an incredibly low R_{dsOn} (only $5.2\text{ m}\Omega$), 35 A of maximum power and a maximum operating voltage of 100 V.

Additionally, it has incorporates shoot-through protection that protects the transistor from a control error, however a dead time needs to be introduced in software. It uses a DFN package which is good for heat dissipation and can always be attached to a heat sink for better performance.

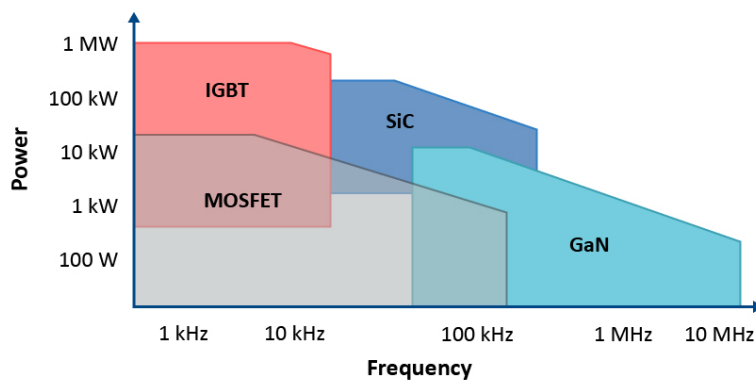


Figure 4.3: Technologies selection based on power and switching frequency [10].

4.2 Processing unit

4.2.1 MCU selection

A Microcontroller Unit (MCU) typically consists of three fundamental components: a central processing unit (CPU), memory, and peripheral modules. The CPU executes program instructions and performs arithmetic and logical operations. Memory is generally divided into non-volatile memory (Flash) for program storage and volatile memory (SRAM) for data and runtime variables. Peripheral modules provide interfaces to the external environment and may include analog-to-digital converters (ADCs), timers, pulse-width modulation (PWM) units, communication interfaces (such as UART, SPI, I²C and CAN), and general-purpose input/output (GPIO) ports. The integration of these components enables standalone operation without the need for extensive external circuitry [33].

When selecting a MCU for a specific application, the first factor to consider is to use an 8, 16, or 32 bit MCU. For this application, a 16-bit MCU would probably be sufficient, however to ensure future-proof and compatibility with other Técnico Solar Boat (TSB) systems, the choice is for a 32-bit MCU. Besides, nowadays, 32-bit MCUs are considered the industry standard since most embedded system Research and Development (R&D) effort is focused on 32-bit cores, and thus both architectures can achieve similar power consumption [34]. There are two main architectures for 32-bit MCUs: AVR and ARM. Being ARM, the most widely used architecture in the industry and the most advanced, it was chosen for this project [35].

STM32 is a family of MCUs that offers a wide range of solutions with different characteristics, making it suitable for various applications, including this project. This family of microcontrollers is used by TSB, so it will be used for this project to maintain the standard.

The MCU requirements for this project are:

- Fast clock frequency: 80 MHz or higher;
- Support 1x CAN peripheral, 1x I²C, 1x UART;
- Able to read 4 or more analog signals;
- 12bit ADC or more.

The final decision was the STM32L476. Although it is not the most suitable STM32 microcontroller for this application, it was selected due to its low cost and because it was already widely used within TSB, reducing development time and associated risks.

The STM32L476 is based on an ARM Cortex-M4 core with floating-point support and operates at frequencies up to 80 MHz. The device provides 1 MB of Flash memory and 128 kB of SRAM, which is more than sufficient for the implementation of the MPPT algorithm, data acquisition, communication protocols, and debugging features. Regarding peripherals, it includes three 12-bit ADCs with up to 16

external channels each, multiple general-purpose and advanced timers capable of high-resolution PWM generation, several communication interfaces (CAN, SPI, I²C, UART, USB), and DMA controllers that allow peripheral operation with minimal CPU intervention.

These features make the STM32L476 particularly attractive for power electronics applications, where simultaneous voltage and current measurements, fast PWM generation, and real-time control execution are required. Furthermore, the presence of a hardware floating-point unit simplifies the implementation of control algorithms and numerical calculations.

If there were no limitations, a microcontroller with a higher clock frequency would have been selected, increasing both the PWM duty-cycle resolution and the available computational power. Additionally, a device with native 16-bit ADCs would improve measurement accuracy and reduce quantization effects. Nevertheless, the STM32L476 provides sufficient performance and peripheral resources to meet the requirements of this application.

4.2.2 Configuration

Since the designed board is custom made, there is any pre-configuration of the MCU in use. In this particular case, the STM32 project needs to be configured in a STM32CubeMX, which is a tool that simplifies the configuration process. In this tool it is defined the Clock frequency, Peripherals working frequency, pin mapping (GPIOs, ADCs, UART, CAN, etc.) and respective configuration, Interrupts, Timers, PWMs, etc. This tool generates a project base on C programming language with all drivers and configurations needed. These files serve as based for firmware development, and can be modified during the project.

Since it is not possible to describe all the configurations used lets highlight the most important ones for this project.

The MCU works at 80 MHz, with an external ceramic resonator of 16 MHz. This frequency is enough for the computational time, and a minimal PWM resolution. It can be flashed using JTAG and it uses an Independent Watchdog (IWDG) time interrupt. The IWDG is a safety feature that resets the MCU in case of a crash or infinite loops. It works by resetting the timer periodically in a specific part of the code, if the timer is not reseted during a specified time period, the interrupt is triaged and the MCU resets automatically.

As for timers, one timer was configured with 100 KHz for PWM and two pins were configured as complementary PWMs with a specific dead time. The dead time was inserted as a safety feature too. The dead time is calculated using Formula 4.26, where the "value" needs to be in the range from 0 to 255.

$$DeadTime = Value / F_{CLK} \quad (4.26)$$

For this project it was tested several values using an oscilloscope. The smallest the value that ensures that both signals are never high at the same time is 2, however 3 was used to add margin. The measured dead time was 32ns, and the rising and falling time were 19ns and 22ns respectively (Figure 4.26).

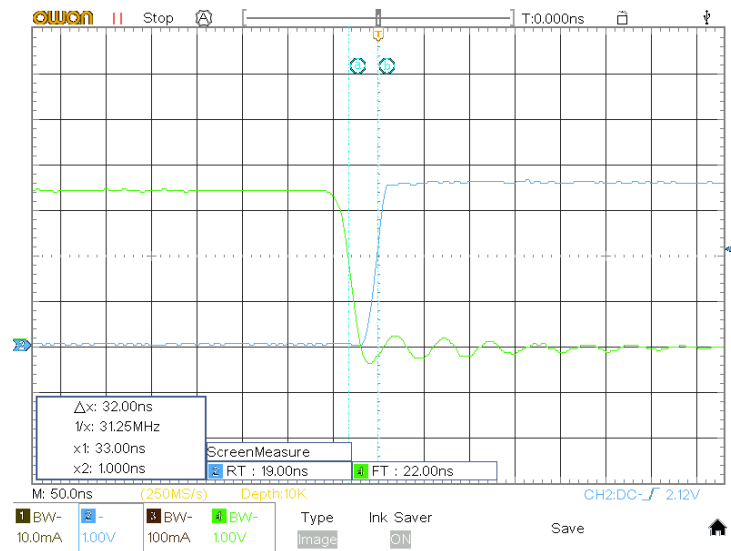


Figure 4.4: Dead time measurement with oscilloscope (value=3).

In simulations the dead time values of 2 and 3 were compared and the difference in efficiency is only 0.01% and therefore the margin is justified.

With 100 KHz PWM and 80 MHz internal clock, the dutty cycle resolution is 0.125 %.

Another timer was configured to trigger every millisecond, and it is used to serve as task interrupts however each task is called at difrent periods (always around milliseconds) which is implemented with a counter and comparators.

The ADC can be configured in a variety of different ways, i chose to use the scan mode which reads multiple inputs in sequence. To read the measured values i am using the pooling method however in theory if the DMA is used instead, the results could be better since it takes less time to read the ADC values and therefore a better average could be made. In practice i could not get reliable results and therefore the pooling method was adopted. The average is made with all data acquired between each MPPT algorithm call. In average, it uses 10 samples.

4.3 Sensors

4.3.1 Voltage and current sensing

Accurate voltage and current measurements are essential for MPPT operation, since tracking algorithms directly rely on these quantities to estimate power and adjust the operating point of the Photovoltaic (PV) array. Measurement noise or inaccuracies directly impact tracking efficiency, especially in high frequency switching DC-DC converters.

Voltage sensing is generally straightforward and can be implemented using resistive voltage dividers followed by ADC. Additionally, it is suggested to use an op-amp as a buffer to reduce the impact of the ADC input capacitor, while charging. This buffer reduces noise and improves the accuracy of the measurement while also allowing the use of higher sampling rates.

Current sensing, however, is more critical and requires careful consideration of the available measurement techniques.

The two main techniques are:

- Hall effect current sensor.
- Shunt resistor sensing with operational amplifiers;

The hall effect current sensor is a non-invasive method that uses the Hall effect to measure current. It provides galvanic isolation and can do bidirectional measurements. However, it is typically more expensive, has lower bandwidth, and may introduce additional noise due to its magnetic sensing nature.

As for the shunt resistor sensing, it is a simple and cost-effective method that uses a low value resistor to convert the current into a voltage drop. This voltage is then amplified using operational amplifiers to match the ADC input range. This method provides high accuracy, low cost, and fast response time, but it requires careful design to minimize power loss since it will dissipate power proportional to the current.

In the prototype a IC based on a shunt resistor sensing method was used, which as a built-in amplifier. The sensor chosen was the INA253A2IPWR, which has a PWM rejection block and small form factor that will be extremely useful for this application. However, this IC is more expensive than a simple shunt resistor and op-amp solution, but it is more accurate and has a better noise rejection. In the future it can be swapped for a lower cost solution, if the performance is acceptable.

Both sensors use the internal ADC from the STM32L476, which has a 12 bit ADC (with 3.0V as the reference). The current sensor output an analog voltage that is equal to the current multiplied by 0.2 V/A. Which means that the resolution of the current measured is $3.0/(4096 * 0.2) = 3.66 \text{ mA}$. As for the voltage sensor, it outputs an analog voltage that is equal to the voltage multiplied by 10/210. The resolution is $21 * 3.0/(4096) = 15.38 \text{ mV}$, which is more than enough for this application.

Both circuits are represented in Appendix 5.3. [place the apendix reference](#)

4.3.2 Temperature and Irradiance sensors

For security reasons it is important to monitor the temperature of the Printed Circuit Board (PCB). Also, the temperature and irradiance of the PV panel are important to monitor for performance analysis and defect detection. These measurements can also be used to improve the MPPT algorithm, since the Maximum Power Point (MPP) is dependent on temperature and irradiance.

For temperature sensing there is a wide range of options, from simple thermistors to more complex digital temperature sensors. For this application a simple thermistor was used, since it is cheap and easy to implement. The thermistor is connected to a voltage divider and the voltage is read by the ADC. The thermistor used to measure ambient temperature and transistor temperature was the ERT-J1VS104FA, which has a resistance of 100kΩ at 25 °C and a B value of 4330K. As for the thermistor used to measure the PV panel temperature, it was the 104JT-025, which has a resistance of 100kΩ at 25 °C and a B value of 4390K. The temperature can be calculated using the Steinhart-Hart equation [36].

$$\frac{1}{T} = A + B \cdot \ln(R) + C \cdot \ln(R)^3 \quad (4.27)$$

This equation uses 3 parameters (A, B and C) that are usually not provided in data sheet but can be calculated using 3 measurements of temperature and resistance. However, the most common approach is to use a simplified version of the equation, which uses only 2 parameters (B and R₀). The temperature can be calculated using the following equation:

$$\frac{1}{T} = \frac{1}{T_0} + \frac{1}{B} \cdot \ln\left(\frac{R}{R_0}\right) \quad (4.28)$$

The thermistor that measures the transistor temperature is also used in an analog protection. This protection uses a comparator to disable the transistors after a temperature threshold that can be modified by changing on resistor in the comparator circuit.

As for the irradiance sensor, the board was designed to use a TSL2591 , which is a digital light sensor with a wide dynamic range and high sensitivity. It can measure light intensity from 0.000118 lux to 88,000 lux, making it suitable for various lighting conditions. The sensor communicates with the MCU via the I2C protocol, allowing for easy integration and data acquisition. However, this sensor is not ideal for outdoor use, since it can easily saturate and his spectrum is not ideal for solar irradiation measurement. The best application for this sensor is in laboratory, however it was not need and therefore it was not used

In the future, if the irradiance value is needed to be used in the MPPT algorithm, a more precise irradiance sensor is needed. The most precise sensors use a PV cell, which based on his temperature, short circuit current and open circuit voltage can provide a good estimation of the irradiance. My suggestion is to design this circuit and use the same cell model that is used in the PV panel to measure the

irradiance. This way, the irradiance measurement will be accurate and with exactly the same spectral distribution as the PV panel which will increase the performance of the MPPT algorithm. This was not implemented in this project due to time constraints, but it is a good option for future work.

4.4 Software

In this section it is explained how does the firmware inside the MCU is structured and how it works. Also, the basic idea behind the communication protocols used and how they are applied to this project as well as the Graphical User Interface (GUI) developed for configuration and data analysis.

4.4.1 Firmware Overview

The firmware is a flag based scheduler with two main task, the MPPT and ADC task.

The ADC task is responsible to convert the analog readings into another units such as current, voltage and temperature. If any of the values exceed the limits configured by the user, a fault is detected and blocks the charging state machine to run during 5 seconds. After this time the ADC task runs again and checks if the faults was removed.

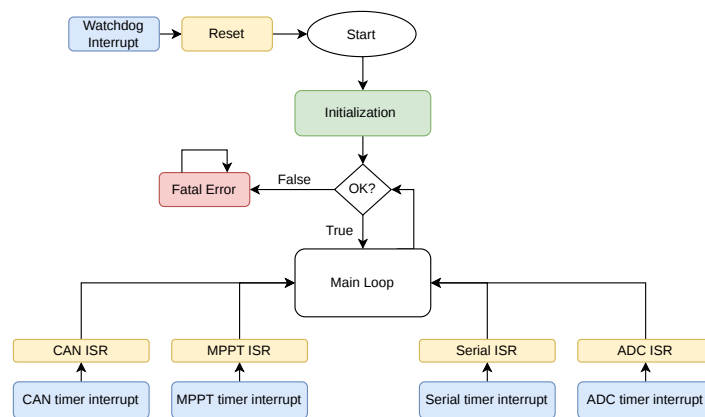


Figure 4.5: Firmware flowchart.

The MPPT task is responsible to compute an average of the last measurements and refresh the state machine which changes the operation mode depending on the measurements and outputs a PWM duty cycle value. In Figure 4.6, is state machine logic.

The sates are divided into charging and not charging. There is 5 "not charging" states:

- **Delay**: waits 100 cycles to skip the transients and then decides the next state;
- **Disable**: disables the DC-DC;
- **No Source**: the solar panel is disconnected and therefore the DC-DC should be disconnected;

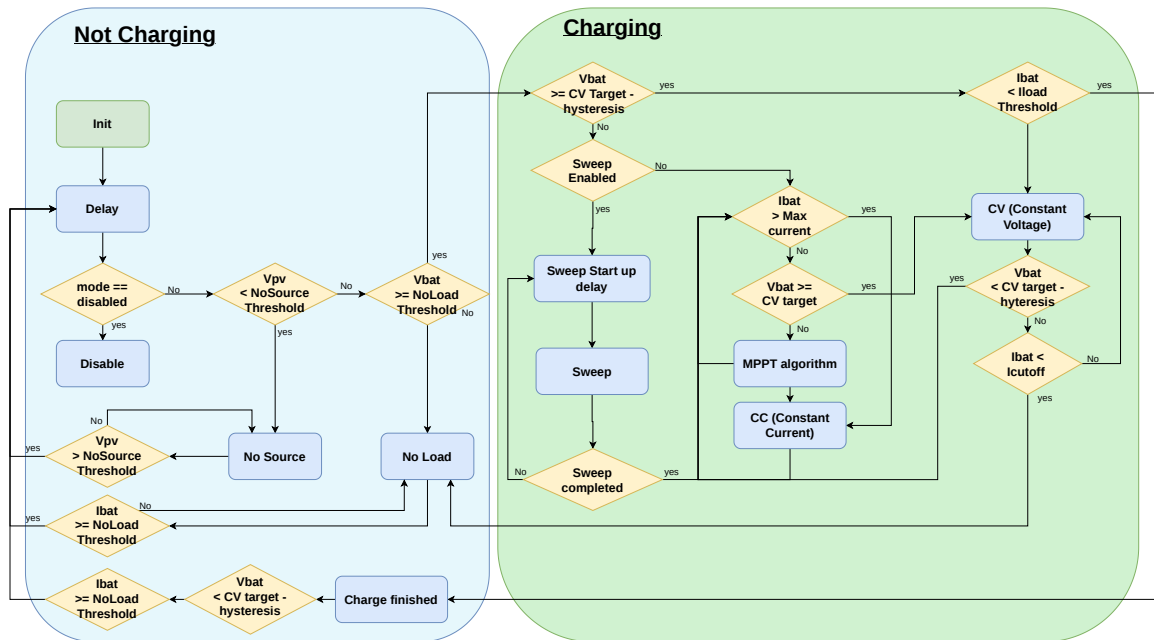


Figure 4.6: Charging state machine state.

- **No load:** the battery is disconnected, and therefore the output voltage is set to the maximum battery voltage to prevent the need of a precharge circuit;
- **Charge finished:** similar to "No load", runs a PID to control the PV side by fixing the battery voltage.

As for the charging states, there is also five:

- **Sweep delay:** wait between sweep steps to ignore transients;
- **Sweep:** decrease the duty cycle until the end of the sweep;
- **MPPT algorithm:** runs the MPPT algorithm chosen;
- **CC:** if the current exceeds the battery limit, runs a PID to fix the current at the maximum value;
- **CV:** runs a PID to control the PV side by fixing the battery voltage.

Besides the main tasks, the board is able to send CAN and Serial messages and therefore the respective tasks.

4.4.2 Serial and CAN communication

One of the most important features of this circuit board is the communication with the rest of the boat. This will enable better decision making relating to energy budget. Also, it will be easier to detect broken or malfunction solar panels or human errors such as bad or forgotten connections.

The main communication protocol is Controller Area Network (CAN). This is the TSB standard protocol and widely use in automotive industry due to its high robustness. The protocol was developed by Bosh in 1986 to let microcontrollers and devices talk to each other without a central host computer.

In the hardware point of view, CAN uses a bus instead of wire to wire communication. Also, it uses twisted wires (CAN High and CAN Low) to block electrical noise. In a software point of view, it uses an error checking system (CRC) ensuring reliability.

For TSB electrical system we developed our CAN protocol on top of the standard CAN communication. Each message is structured as shown in Figure 5.3 following the CAN 2.0A standard.

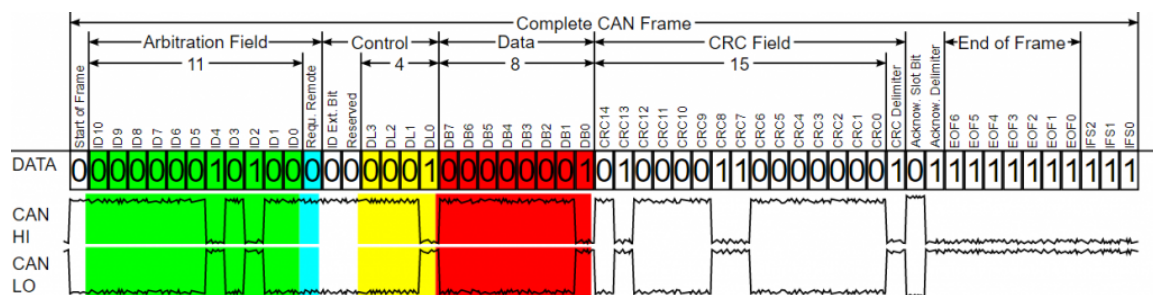


Figure 4.7: Dashboard.

The 11 bits marked in green (Arbitration Field) can be used by each protocol differently. Usually, it is used to send priority bits, device ID and message ID. In the current vessel (SG01), the priority bits were removed and used to increase the number of devices on bus. With the current structure (Tab 4.3), the protocol supports up to 32 devices with 63 messages each.

Table 4.3: TSB CAN messages Arbitration Field.

Message ID	Device ID
6 bits	5 bits

In Control field, the first bit is to use Extended IDs and next one is reserved. The yellow field is called Data Length Code (DLC) and is where the length of the data is defined. The maximum length is 8 bytes. Then there is the Data field (represented in red).

The rest of the fields are treated by the CAN library which is an adaptation of the open source library called STM32_CAN by pazi88.

For the MPPT it was defined an ID structure, where the device ID is maintained intact, and Message ID is divided in two. The 3 most significant bits represent the MPPT number from 0 to 7 and the rest of the bits identifies the messages. This structure gets enough message IDs for each MPPT as well as not occupying a lot of Device IDs. In Table 4.4 is showed an example of the corresponding IDs when the Device ID is 0x0D (default ID).

The Device ID can be configured/changed via GUI however the logic is maintained.

Table 4.4: CAN IDs for each MPPT number and message.

MPPT Number	MSG:MPPT_bat	MSG:MPPT_pv	MSG:MPPT_aux1	MSG:MPPT_aux2
0	0x00D	0x02D	0x04D	0x06D
1	0x10D	0x12D	0x14D	0x16D
2	0x20D	0x22D	0x24D	0x26D
3	0x30D	0x32D	0x34D	0x36D
4	0x40D	0x42D	0x44D	0x46D
5	0x50D	0x52D	0x54D	0x56D
6	0x60D	0x62D	0x64D	0x66D
7	0x70D	0x72D	0x74D	0x76D

The CAN protocol is excellent for communication between boards however it is not the most practical protocol for configuration because the CAN IDs can change between vessels and implies the use of an external hardware to communicate between the board and a computer.

In contrast, serial communication is done by just connecting a wire to the board and since the protocol is only used for configuration and data analysis it can be the same forever and if it needs to be changed, it does not depend on any vessel.

There is no default protocol to send data over serial, however during the Battery Management System (BMS) thesis a serial protocol was created by the team and will serve as a basis for this project. Unfortunately the firmware library is not compatible with STMs, so it was created from scratch.

The protocol uses the structure defined in Table 4.5. It begins with the Start Byte which is defined as 0xAC, followed by the device ID which defines the source of the message and the message ID defining what message is the device sending, the data length, the data, checksum byte and the end byte (0xAD) marking the end of the message.

Table 4.5: TSB Serial messages format.

Start	Device ID	MSG ID	LEN	Data	Checksum	END
1 Byte	1 Byte	1 Byte	1 Byte	8 Bytes	1 Byte	1 Byte

Similar to CAN protocol, for each message it was defined a Message ID respecting the same logic. This ID alongside with the Device ID and length is initialized in the beginning of the code, the rest of the variables are treated by the library except the data field that is changed before sending any message.

The Data of each message is structured in the same way as in CAN protocol however it was added 4 messages for configuration. These messages enable the change of current limits, voltage limits, MPPT parameters, CAN settings and CC-CV targets.

4.4.3 EEPROM Emulation

To save the configuration received from the serial protocol, a library was used that emulates an EEPROM which is a type of non-volatile memory that retains its contents after power down. EEPROMs have been

used for many years in a wide range of devices to store settings, calibration data, and backups. In this project, no external EEPROM was added to the board, and the MCU does not include one internally. However, the library can use internal flash memory, which is normally reserved for program code, to store data, thereby emulating EEPROM behavior.

The main disadvantages of this approach are limited storage capacity and reduced write endurance compared with a dedicated EEPROM. The STM32 internal flash can typically withstand on the order of 10,000 erase cycles per page, whereas external EEPROM devices often support roughly one order of magnitude more. Even so, this is not a practical limitation in this application. Configuration data is stored as a small set of 16-bit variables, and each save operation appends new records to flash rather than overwriting the previous value in place. Each page can hold 252 variables. For this project, a full board configuration requires 13 write operations, therefore, one flash page can hold many configuration updates before it becomes full and must be erased. Furthermore, the emulation is configured to use ten flash pages, with wear levelling implemented through page transfer and cleanup. When a page is full, the latest values are copied to another page and the old page is eventually erased. This distributes erase cycles across multiple pages and extends the effective lifetime of the memory.

In conclusion, neither storage capacity nor endurance is expected to be a problem for this project. Under the current configuration, a complete configuration upload would need to be performed hundreds of thousands of times before the theoretical endurance limit of a flash page is reached, which is far beyond the expected use case of occasional configuration updates over serial.

Since the emulated EEPROM shares the same physical flash as the firmware, care must be taken to ensure that program code is never placed in the reserved EEPROM region. This was achieved by modifying the linker script, which defines the memory map used during the build process. The length of the program flash region was reduced so that firmware is linked only below the first address saved for EEPROM emulation.

4.5 Graphical User Interface

In parallel with the design and test of the MPPT board, a GUI was designed to make this project more accessible and easy to use. The application was built from scratch, however it was inspired in the application built during BMS thesis. It was built using QT framework, which is a powerful, cross platform software toolkit used to build high performance applications and GUIs that run on multiple operating systems like Windows, macOS, Linux.

This GUI communicates with the board with Serial or CAN communication. To use CAN communication, the computer should be connected to the bus with a tool called PEAK-CAN.

After connecting the computer to the board, the application is able to identified which board is con-

nected and show it to the user in the Dashboard page, Figure 4.8. In the main page the user selects which board he is working on. The application is prepared for several PCBs, however only the BMS and MPPT are correctly working. The rest are placeholders for future improvements.

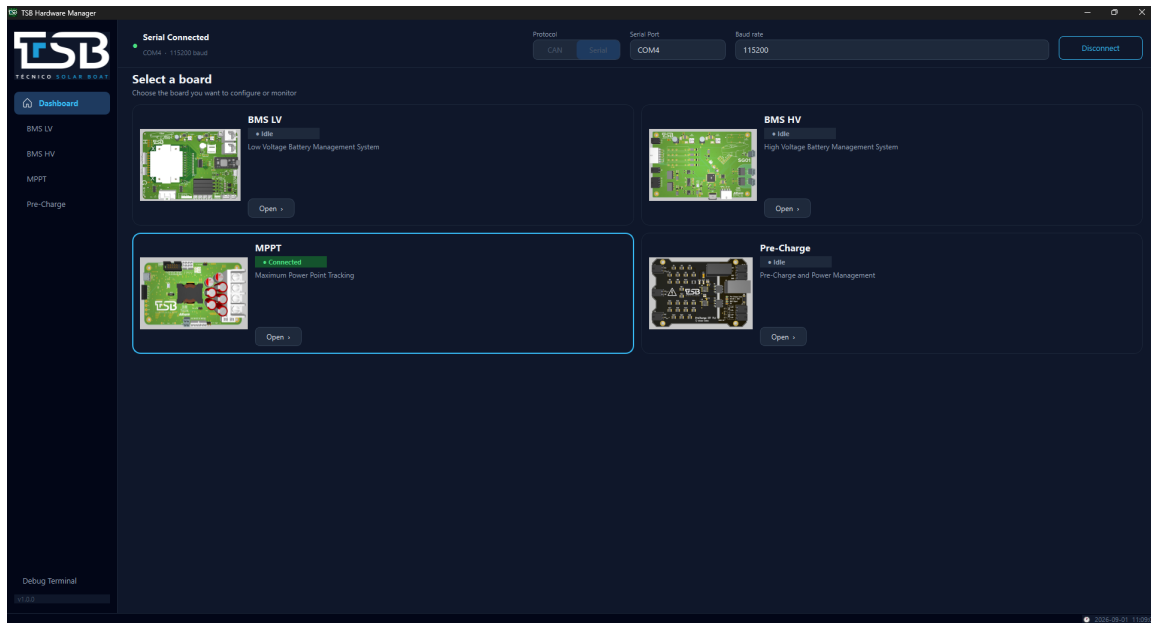


Figure 4.8: Dashboard page.

When the MPPT page is selected (Figure 4.9), a new tab opens showing the most important variables, such as voltage, current, power temperature, efficiency, irradiance and duty cycle. Additionally, there are two visual representations. On the left there is a P-V graph with theoretical curves for different irradiance values and a blue dot placed on the current point of operation and updated with the received data.

On the right, it is represented the CC-CV curve, where the received battery side data is represented over time.

On top of the page, the user can switch to the configuration tab (Figure 4.10). In this tab the user can read the configuration currently installed on the board, reset this configuration or upload a new one.

The last page is the debug page (Figure 4.11), where the user can visualize both Serial protocol messages and serial prints at the same time.



Figure 4.9: MPPT tab. **atualizar**

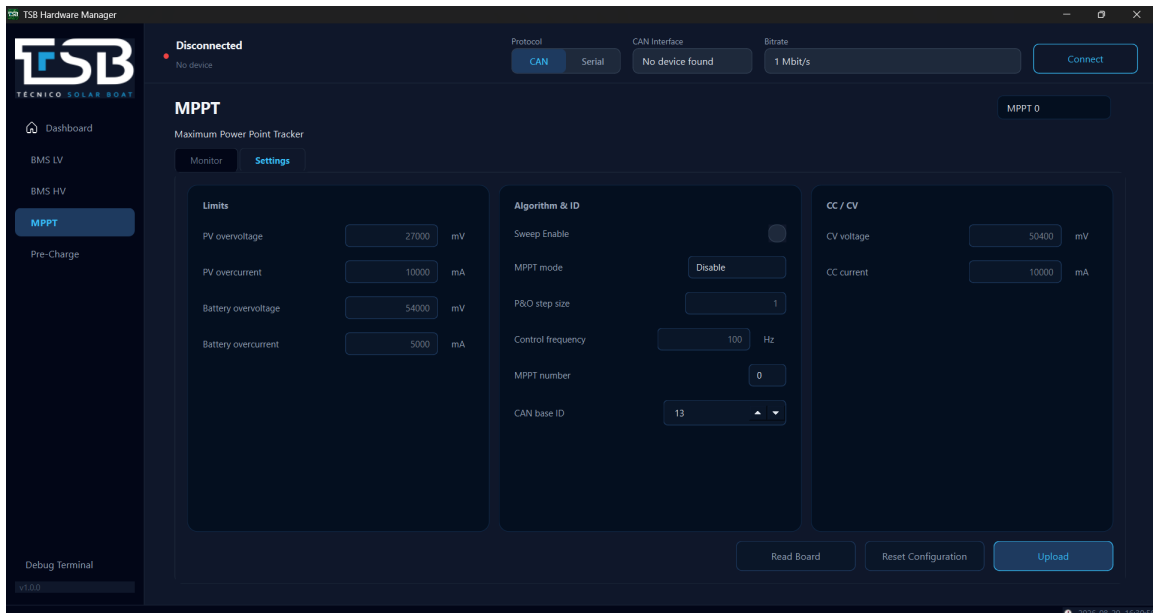


Figure 4.10: MPPT configuration tab.

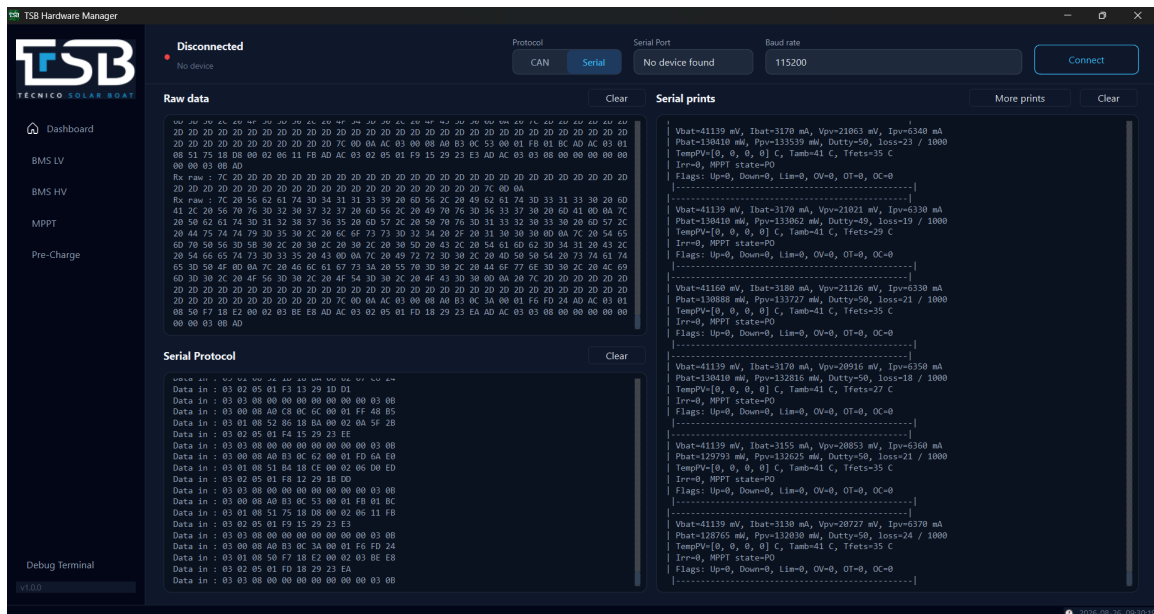


Figure 4.11: Debug page.

5

Simulations

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Simulations were used to validate the converter design and the Maximum Power Point Tracker (MPPT) algorithm before hardware implementation. This chapter presents the LTspice simulations of the photovoltaic model and the boost converter, an analytical power loss estimation in MATLAB, as well as the evaluation of the tracking algorithm in Simulink.

5.1 DC-DC Converter

The converter was simulated in LTspice to confirm the component selection under representative charging conditions. This section presents the photovoltaic cell model and the boost converter simulations used to compare inductor and capacitor options and to estimate ripple and efficiency.

5.1.1 Solar Panel model

The simulations started at LTspice with the aim to test the converter at different conditions. To achieve that, a Photovoltaic (PV) electrical model of the cells were designed using a five parameter model (1M5P) configuration. There are more electrical models that can simulate a solar panel, some with more accuracy but more complex. This model was perfect for the aim of the simulation since it is both reliable and simple.

The 1M5P model represents the solar cell through a single diode equivalent circuit composed of a current source in parallel with one diode, a series resistance R_s , and a shunt (parallel) resistance R_{sh} , as shown in Figure 5.1. The current source represents the photo generated current I_{ph} , which is proportional to the incident irradiance, while the diode accounts for the non linear I - V characteristic of the p-n junction. The series resistance models the losses associated with the material bulk resistance, metal contacts and interconnections, whereas the shunt resistance represents the leakage currents through the cell, typically caused by manufacturing defects.

The output current of the model is given by the characteristic equation 5.1, where I_0 is the diode saturation current, n is the diode ideality factor, and V_t is the thermal voltage [37]. Together with I_{ph} , R_s and R_{sh} , these constitute the five parameters that give the model its name.

$$I = I_{ph} - I_0 \left[\exp \left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad (5.1)$$

These parameters are not directly provided by manufacturers, but can be extracted from the standard datasheet values, namely the open-circuit voltage (V_{oc}), short-circuit current (I_{sc}), voltage and current at the Maximum Power Point (MPP) (V_{mp} , I_{mp}), and the respective temperature coefficients, through a set of equations that ensure the model matches these reference points under Standard Test Conditions (STC). There are several ways to do it, however it is outside the range of this work. To simplify,

calculations were carried using online tools and will not be explained in detailed here.

This approach allows the panel behavior to be accurately reproduced under different irradiance and temperature conditions, while keeping the circuit implementation simple enough to be integrated directly into LTspice.

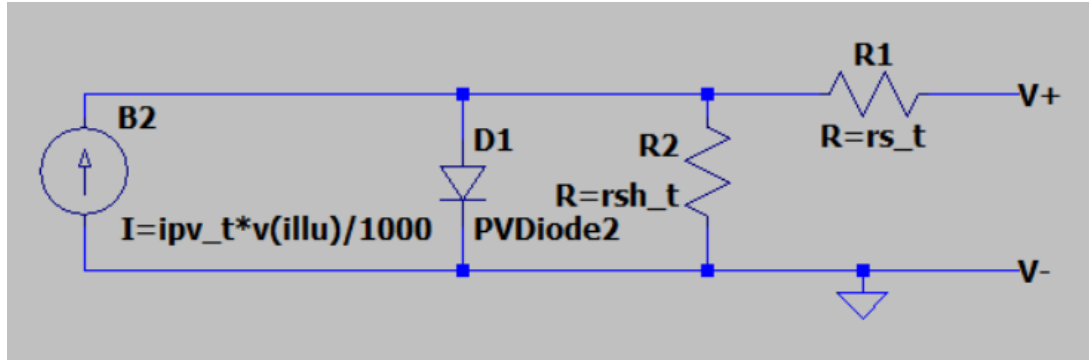


Figure 5.1: Solar panel model diagram (1M5P configuration).

5.1.2 Boost Converter

To validate the theoretical calculations, simulations in LTspice were carried using the boost converter in steady state mode and a fix duty cycle.

Considering a solar panel operating at his MPP with 35 cells, five points of CC-CV charging curve were chosen to test the performance of the converter. These five points are represented in Figure 5.2.

It is important to note that in this CC-CV curve the constant current is given by the solar panel and not by the battery as it is usually expected. This is only truth due to the maximum charge current of the battery being lower than the current that can be provided by the solar panel. So at CC, the converter is working at MPP. Also, the timescale will depend on the battery capacity.

With the chosen five points, some simulations were carried using capacitors and inductors that would fit the requirements calculated earlier. The objective of these simulations is to optimize the performance of the converter, leading to better choices.

The model was built in LTspice with approximated models given by the manufactures for higher precision. Only the solar panel model was designed by hand, as previously mentioned.

The battery was simulated using a voltage source and a resistor representing the internal resistance of the battery.

The dead time of the transistors was implemented too. Using laboratory measurements, the rising and falling timing was measured and used in simulation. Has for the dead time, it is used the same as in the final prototype.

The first simulations were carried to choose the input and output capacitors. The idea is to simulate

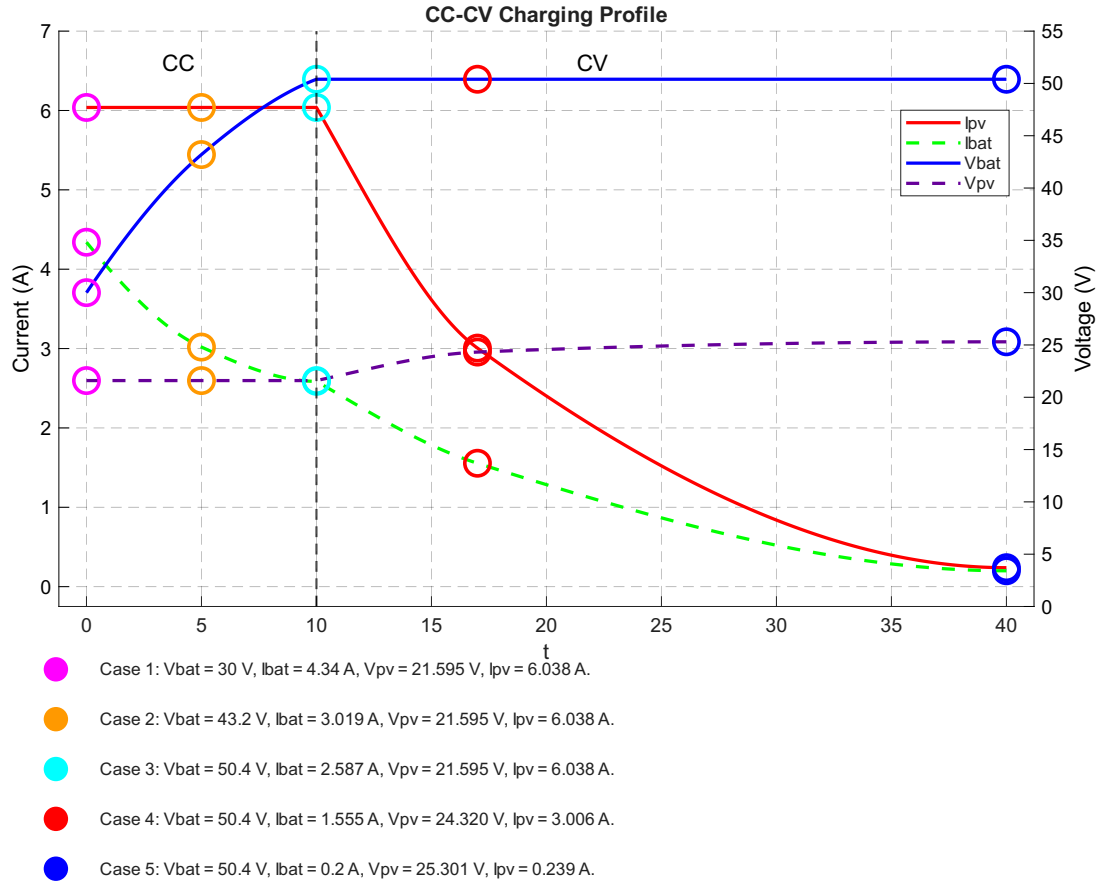


Figure 5.2: CC-CV charge characteristics and analysis point selection.

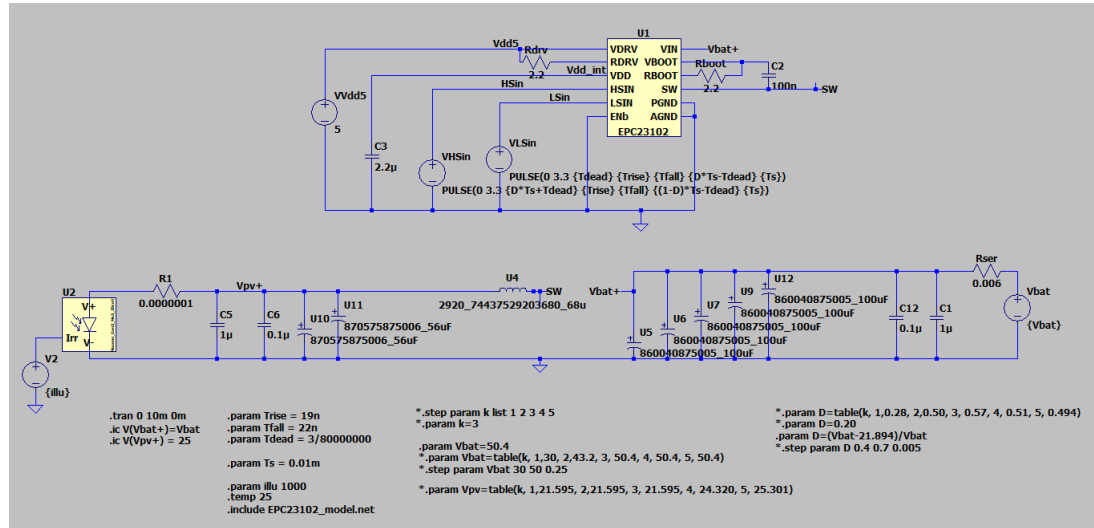


Figure 5.3: Simulation circuit schematic.

the circuit using a known inductor (68 uH) and change the capacitors for different configurations. All configurations are overdimensioned to give low ripples (current and voltage). So the main factor is the

efficiency of the converter.

In Table 5.1 is shown the results. The efficiency of all combinations are very high however the highest is the one with two 56 μF as input capacitors and tree 220 μF capacitors as output capacitors. Unfortunately the 220 μF were not available at the moment, so the second best option was chosen, the five 100 μF capacitors.

It is worth noting that this is a simulation and therefore the efficiency is higher than what will be measured in laboratory results. Additionally, the difference between each combination it is very small. To make the product cheaper the third option would be as great as the one chosen.

Table 5.1: Loss (%) for different capacitor configurations (C_{in} , C_{out}).

Case #	2x56 μF , 3x220 μF	100 μF , 3x220 μF	100 μF , 5x100 μF	2x56 μF , 5x100 μF
1	0.5729	0.5796	0.5856	0.5788
2	0.6762	0.6985	0.7069	0.6844
3	0.7359	0.7655	0.7743	0.7449
4	0.6654	0.7091	0.7161	0.6735
5	3.4177	3.7148	3.7554	3.4671

Similar to the previous simulation, the inductor was chosen by keeping all components with the same values (the capacitors used were $C_{in} = 2 \times 56 \mu\text{F}$ and $C_{out} = 5 \times 100 \mu\text{F}$) and changing the inductor. Similar to before, all inductors simulated are within the minimum required.

The losses for each case are represented in Table 5.2. Similar to before, the efficiency difference is very low but not as linear as in the previous simulation, likely due to the inductor models chosen.

The inductor chosen will highly influence the input ripple which can generate heat, and electrical interference. Table 5.3 shows the different ripple currents for each inductor in each case of the CC-CV curve. The 22 μF inductor showed higher ripple current reaching values close to the input current. The inductor chosen was the 68 μF since it has overall good efficiency and reasonable input current ripple.

Table 5.2: Loss (%) for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	0.4206	0.4439	0.5453	0.5788	1.2659
2	0.6645	0.6793	0.6627	0.6844	1.3664
3	0.7883	0.8076	0.7290	0.7449	1.4193
4	1.0556	1.0583	0.6867	0.6735	0.9624
5	6.7045	6.6230	3.6556	3.4671	3.8177

Now that the converter components are all chosen, the following measurements were taken in LT-spice:

Both voltage ripples are lower than what was defined as the goal in Table 4.2. As for the current ripple it is still reasonable. The losses/efficiency are also higher than what was aimed, reaching efficiency values up to 99.4%.

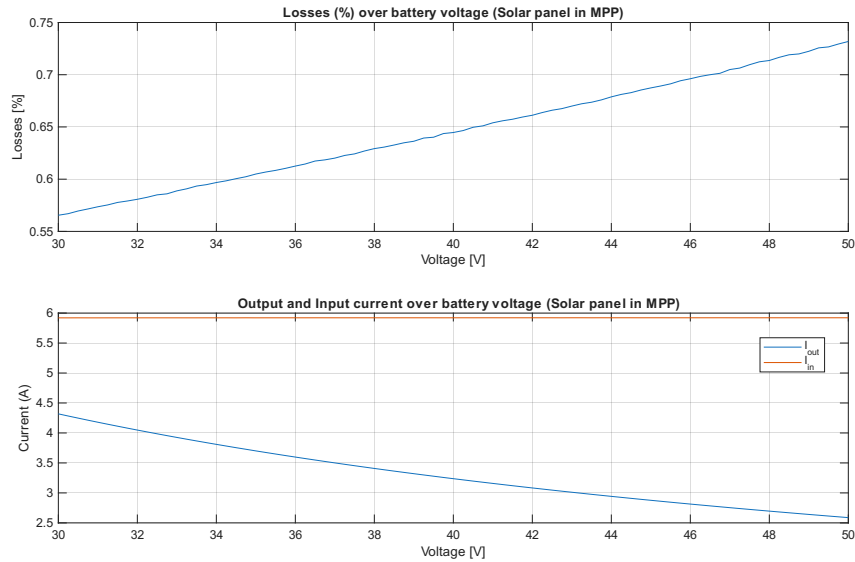
Table 5.3: Δi_L ($I_{L,\max} - I_{L,\min}$) for different inductance values.

Case #	22 μH	33 μH	47 μH	68 μH	100 μH
1	2.6939	1.8906	1.3256	0.9781	0.6251
2	4.7934	3.3416	2.3354	1.7066	1.0707
3	5.4733	3.8253	2.6813	1.9644	1.2839
4	5.5596	3.8565	2.7075	2.0139	1.2925
5	5.5520	3.8510	2.6903	1.9636	1.2344

Table 5.4: Ripple and power loss summary for each operating point.

Case #	ΔV_{bat} (V)	ΔV_{pv} (V)	ΔI_L (A)	Loss (%)
1	0.0406	0.0145	0.9781	0.578
2	0.0491	0.0212	1.7066	0.684
3	0.0529	0.0244	1.9644	0.745
4	0.0406	0.0220	2.0139	0.674
5	0.0302	0.0215	1.9636	3.464

A voltage sweep were performed, and as expected the efficiency drop with the increase of output voltage (Figure 5.4). Also, as mentioned earlier, the output current drop because in Constant Current (CC) mode the solar panel current is constant, so the output current drops with the increase of battery voltage due to the power conservation principle.

**Figure 5.4:** Battery voltage sweep simulation results.

Note: in version 1.1 of the prototype it was added two shotcky diodes that were not considered in these simulations. The diodes slightly increase the efficiency of the converter (+0.03%) due to their lower forward voltage. The increase could be higher if the dead time was longer. Since the dead time is only 0.375 % of the PWM period, the difference is negligible and therefore its use is irrelevant.

5.2 Power Loss Estimation

Although LTspice provides an accurate estimation of the converter efficiency, it does not directly identify the contribution of each component to the total power dissipation. Therefore, an analytical power loss estimation was developed in MATLAB to estimate the individual losses of each component of the proposed synchronous boost converter. This estimation uses the electrical parameters of the selected components obtained from their datasheets together with the converter operating conditions.

The total prototype losses are obtained by summing the losses associated with the GaN transistors, inductor, capacitors and auxiliary electronics:

$$P_{loss} = P_{tran} + P_L + P_C + P_{aux}. \quad (5.2)$$

The prototype efficiency is then calculated as:

$$\eta = \frac{P_{out}}{P_{out} + P_{loss}} \times 100\%. \quad (5.3)$$

The transistor losses can be split in three. The conduction, switching and deadtime losses:

$$P_{tran} = P_{cond_{HS}} + P_{cond_{LS}} + P_{sw_{HS}} + P_{sw_{LS}} + P_{dead} \quad (5.4)$$

The conduction losses of each GaN transistor are given by:

$$P_{cond} = I_{RMS}^2 R_{DS(on)}. \quad (5.5)$$

These losses are primarily caused by the on state resistance of the transistors, resulting in power dissipation whenever current flows through the channel.

The switching losses are estimated using Equation 5.6, which takes into account the losses during the period when the transistor is not fully on nor fully off. Also, the losses associated with charging and discharging the output capacitance of each transistor.

$$P_{sw} = \frac{1}{2} V_{out} I_{L_{AVG}} (t_r + t_f) f_{sw} + \frac{1}{2} C_{oss} V_{out}^2 f_{sw} \quad (5.6)$$

The deadtime losses are calculated using Equation 5.7. These losses originate from the conduction of the reverse conduction path during the dead time interval required to prevent shoot through. This reverse condition path is either from the third quadrant conduction from the GaN device or from the parallel schottky diode added later in version 1.1 of the prototype.

$$P_{dead} = 2 \cdot V_f \cdot I_{L_{AVG}} \cdot t_{dead} \cdot f_{sw}. \quad (5.7)$$

The inductor and DC capacitor losses are estimated using the RMS current passing through them and their Equivalent Series Resistance (ESR). The inductor manufacture does not provide the required magnetic material parameters to estimate the AC losses. However, an employee suggested using their own tool called REDEXPERT which can simulate both DC and AC losses. Therefore, the value AC losses were copied from their tool.

$$P_L = I_{L_{RMS}}^2 ESR + P_{L_{AC}}. \quad (5.8)$$

$$P_C = I_{C_{RMS}}^2 ESR, \quad (5.9)$$

The gate driver losses are estimated using Equation 5.10 which takes into account the energy needed to drive the transistors. Although the EPC23102 has an internal gate driver, this is still relevant.

$$P_{gate} = (Q_{g_{LS}} + Q_{g_{HS}}) * V_{drv} * f_{sw}. \quad (5.10)$$

Last but not least, the rest of the electronics (MCU, DC-DC, sensors, etc.) were considered too. A sum of the maximum current of every component was made. Then the power losses can be estimated based on the power and efficiency of each DC-DC converter, Equation 5.11.

$$P_{aux} = (I_{3v3} * 3.3) * eff_{dc} + (I_5 * 5) * eff_{dc} + I_{in} * R_{shunt} + I_{out} * R_{shunt} + P_{gate} \quad (5.11)$$

Figure 5.5 presents the resulting power loss distribution. The calculated total power loss is approximately 1.98 W, corresponding to an efficiency of 98.50%. The efficiency dropped around 1%, compared to LTspice simulations and it is expected to drop even further in real laboratory results since the Printed Circuit Board (PCB) will generate losses too.

The largest contributors to the total losses are the DC losses of the inductor and the ESR losses of the capacitors. This highlights the importance of selecting an inductor with low winding resistance and capacitors with low ESR, as these passive components represent a significant portion of the converter losses.

The auxiliary electronics also have a considerable impact on the overall losses, primarily due to the losses associated with the DC-DC converter.

The conduction losses of the GaN transistors account for approximately 24.4% of the total losses, further highlighting the importance of selecting transistors with low on-state resistance.

Additionally, the interconnections and traces on the PCB exhibit parasitic inductance, which con-

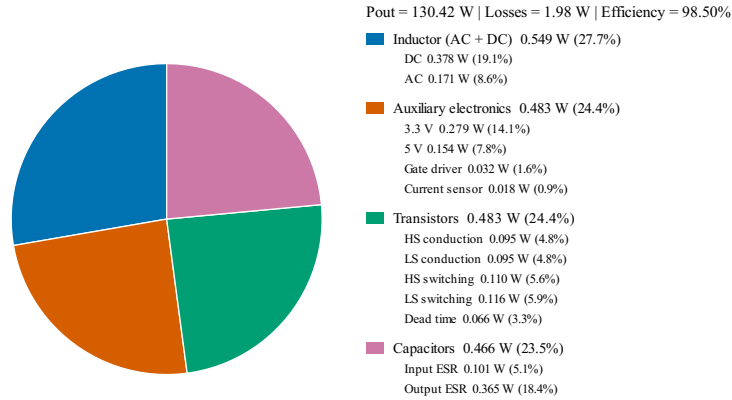


Figure 5.5: Prototype analytical power loss distribution at Case 2.

tributes to the overall losses of the device, which were not taken into account in this analysis.

Overall, the analysis indicates that nearly half of the converter losses originate from the passive components, while the semiconductor devices account for approximately one quarter of the total losses. Nevertheless, both the simulation results and theoretical analysis indicate that the selected components and design approach result in a highly efficient converter.

5.3 MPPT algorithm

The algorithm has a few parameters like the working frequency, sampling frequency, step size, power dead band, etc. All of this parameters need to be tuned to work on his full potential while keeping it safe. Therefore, a simulation in MATLAB Simulink was implemented which simulates the behavior of the switching converter with the values defined earlier. The boost converter is connected to a solar panel and a battery, as it is spouse to be use (Figure 5.6). To simplify, the simulation uses a diode instead of a transistor making it a basic boost converter. This simplification will not have influence on the results since the idea is to analyze the performance of the algorithm instead of the converter efficiency.

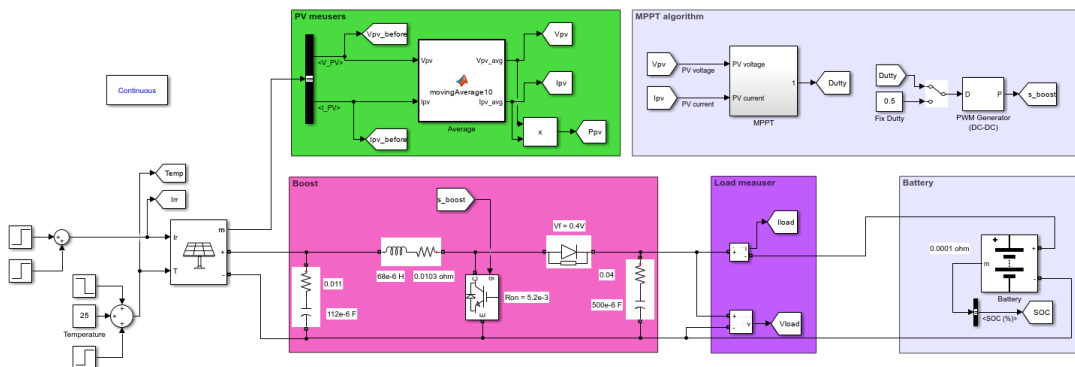


Figure 5.6: MATLAB Simulink MPPT simulation model.

The solar panel and battery are both configured with the parameters of the real system. It was used the specifications of solar panel with 35 cells and the São Gabriel 01 (SG01) battery. The PV side sensors are inside the solar panel and show some noise as in the real system, to mitigate it, an average of 10 samples was implemented with each sample collected at 1 KHz. The value of the average is sent to the algorithm which runs at 100 Hz. Inside the MPPT block are the sweep algorithm and Perturb and Observe (P&O).

To understand the impact of each environment variation on the MPP point, a sweep was performed for different irradiance values and temperatures, Figure 5.7 and 5.8. For irradiance variations, the maximum duty cycle difference is 3%. On the other hand, the maximum duty cycle difference for temperature variations is 9%. This means that the algorithm will always be slower to temperature variations when compared to irradiance variations. However, it is more common to have a fast changing irradiance than temperature. The solar panels will heat slowly and when covered in water they cool around 10 to 20°C (values measured during tests).

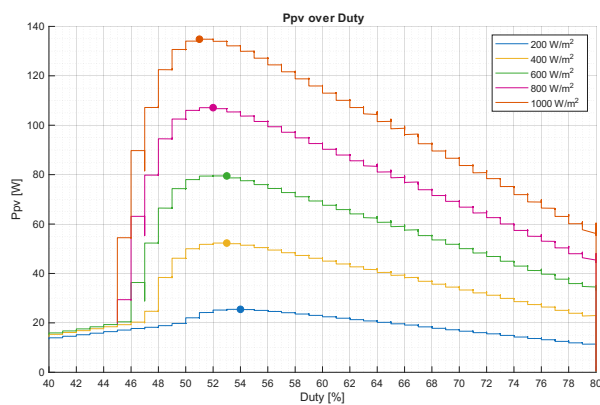


Figure 5.7: Duty cycle sweep with different values of irradiance.

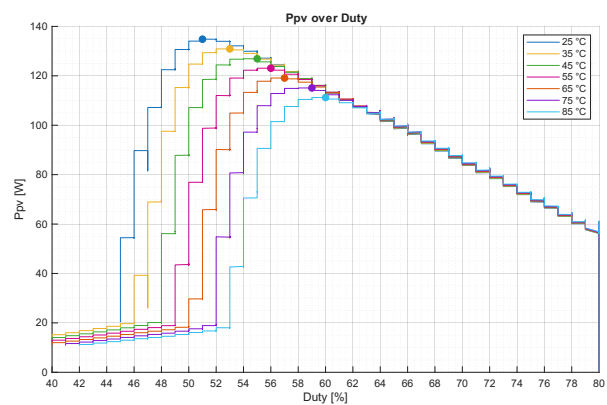


Figure 5.8: Duty cycle sweep with different values of temperature.

Let's consider the worst case which is 9%. To achieve the desired speed of 0.5 seconds, a combination of the step size and working frequency needs to be tuned. For example, considering a working frequency of 100 Hz, the algorithm have 50 steps to achieve MPP under 0.5 seconds. So if the algorithm uses a step size of 9% divided by 50 (0.18%), in an ideal situation it would reach the speed goal every time. However, since this is a small step it can be confused with noise. Therefore, always needs to be tuned to the systems where is being applied.

In simulations, the system can be tested with different parameters to give an idea of how it should behave outside simulation. Fixing the frequency to 100 Hz, a simulation was performed with 3 different step sizes under a few steps of temperature, Figure 5.9. The smaller step is too small, and therefore it never reaches the MPP. Most likely due to the noise. The other two show a similar performed, but the higher step show more oscillations around the MPP.

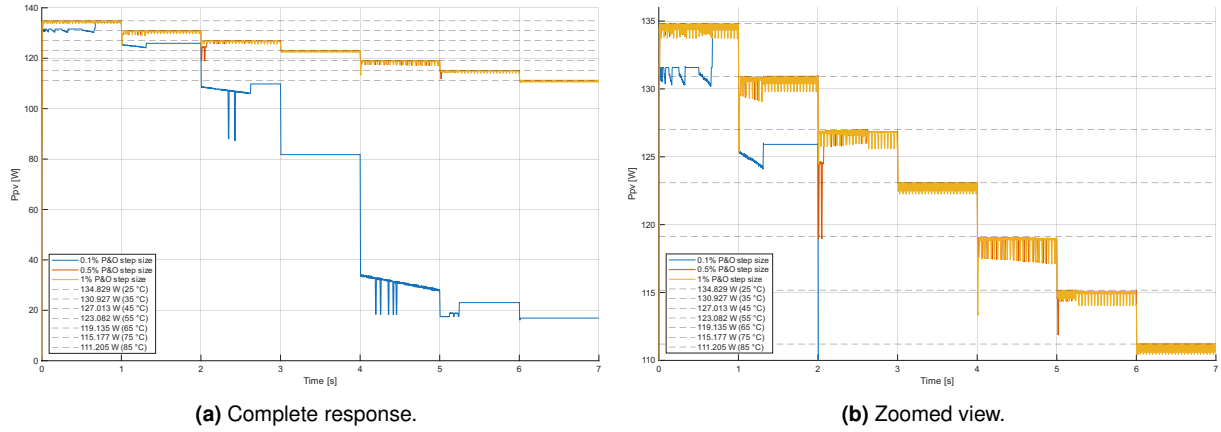


Figure 5.9: Power response for different P&O step sizes during a temperature variation of 10 °C per second.

The analysis was then extended to more severe operating point variations. For a temperature step from 25 °C to 85 °C (Figure 5.10), the 0.5% duty-cycle step, which had performed well in the previous test, required the longest tracking time, approximately 1 s to reach within 5% of the MPP. Given that the tracking time requirement is at most 1 s, and typically below 0.5 s, this step size lies at the limit of the specification. The 1% and 2% steps reached the MPP faster, but with larger oscillations around the operating point. A step size between 0.5% and 1% is therefore considered the most suitable compromise.

It should also be noted that the algorithm stalled on a few occasions. This behavior is attributed to measurement noise and to the P&O deadband. Although not desirable, it is consistent with the expected operation of the method and can be reduced by further tuning.

A rapid irradiance variation was also simulated. As shown in Figure 5.11, the MPP is reached almost instantaneously for all step sizes under consideration. This result is expected, as discussed earlier, since the corresponding duty cycle variation is approximately three times smaller than in the temperature case.

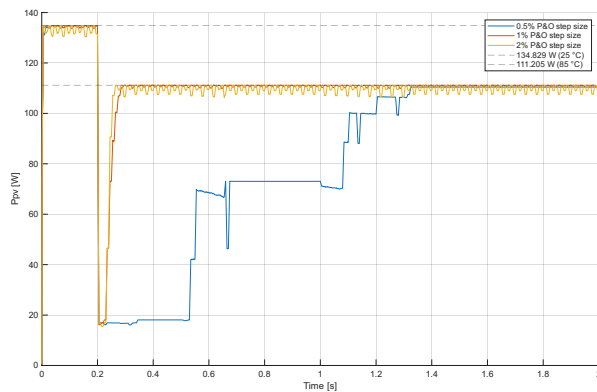


Figure 5.10: Power response for different P&O step sizes during a temperature variation from 25°C to 85°C.

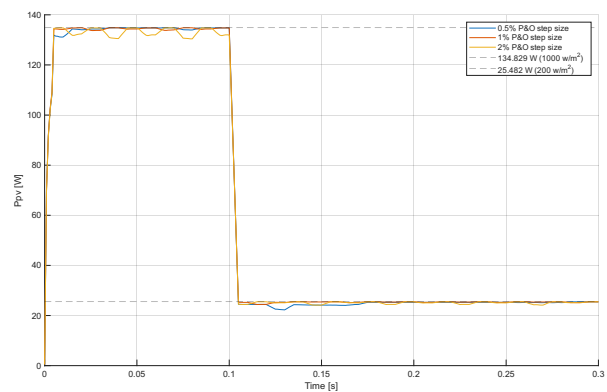


Figure 5.11: Power response for different P&O step sizes during a Irradiance variation from 1000 W/m² to 200 W/m².

6

Laboratory results

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Following the design and simulation stages, a prototype was manufactured and assembled. However, practical implementations may differ from simulation results, as simulations generally rely on idealized component models and do not fully account for parasitic effects associated with the components and the Printed Circuit Board (PCB). Therefore, laboratory testing was conducted to experimentally validate the design and verify that the prototype operates as expected.

6.1 Design and assembly

All boards developed as part of this project, shown in Figure 6.1, were designed using Altium Designer, manufactured by Eurocircuits, and assembled by hand to minimise production costs. The total cost of the electronic components is approximately 90 €, although this value may vary depending on the quantities purchased. The PCB manufacturing cost ranges from approximately 3 € to 35 €, depending on the manufacture.

Consequently, the total cost of a complete prototype is approximately 125 €, which is lower than the cost of the product currently in use. Furthermore, the effective cost can be further reduced through component and manufacturing sponsorships.

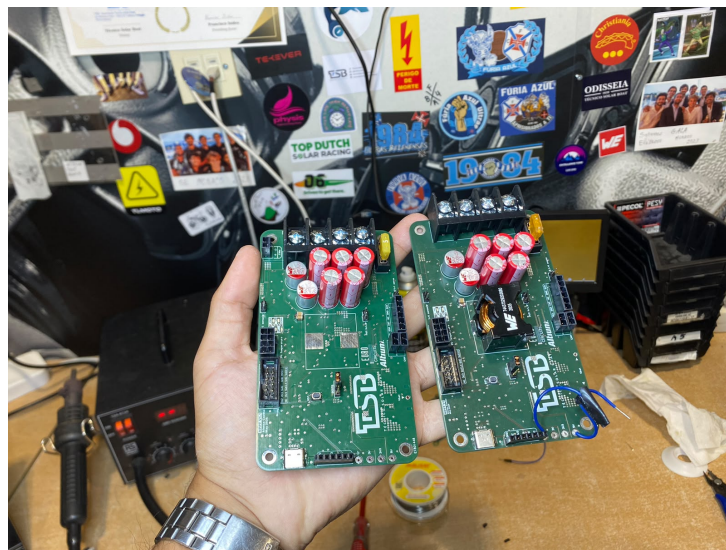


Figure 6.1: First and second iteration side by side. [place a better image](#)

The **first prototype (V1)** iteration presented several issues that were identified during the initial testing phase. For instance, an incorrect 3D model resulted in the main connector being positioned incorrectly (Figure 6.1). Additionally, the flash connector did not provide 5V which made test more difficult. The UART signals connected to the usb-c connector were swapped, and therefore the communication was only possible using a external USB to UART adapter.

Although none of these issues were critical for the fundamental operation of the circuit, they made the testing phase more time consuming and complex. Therefore, a second iteration was designed and manufactured with all the issues fixed.

Besides the errors found in the first prototype, the **second iteration (V1.1)** two schottky diodes were placed in parallel with the boost transistors which accordingly to the datasheet of the transistors, should improve the efficiency of the boost converter. After some simulations, it was found that this statement does not apply in this case due to the short deadtime. Furthermore, during testing of the first iteration, an increase in the inductor temperature was observed. To address the possibility of requiring additional cooling, a connector was added to place a fan if needed. These will also be analyzed in this section.

Near the end, a **final version (V1.2)** was made, which includes the extra filter capacitors to filter noise which will be discussed later. Additionally, the sensors filters were placed near the sensors. Since switching devices generate electromagnetic noise, the filters should be close to the Microcontroller Unit (MCU) pins.

Due to time constraints, the majority of the results presented in the following sections were obtained using version V1.1. Whenever results obtained with the latest version are presented, this will be explicitly stated, otherwise, the results refer to version V1.1.

6.2 Voltage and Current sensor

Both current and voltage sensors were tested in the laboratory to validate the performance and accuracy of the measurements.

The voltage sensor test was performed by applying a known voltage to the input and measuring MCU readings.

The test was performed over the entire voltage range required for the application, from 0 V to 51 V. The results demonstrate that the voltage sensors provide accurate measurements, with average absolute errors of 60 mV for the battery voltage measurement and 112 mV for the solar panel voltage measurement. The error is different for each sensor because it depends on the actual resistance of the resistors used in the voltage dividers and the PCB layout. For each board the error will be different, but it is expected to be in the same order of magnitude since the tolerance of the resistors are equal.

These could be improved even further by using resistors with less tolerance and lower temperature coefficient or by increasing the ADC resolution. Another option that does not increase the cost of the system is to subtract the linear error of the results. The magnitude of the error is small, and therefore it does not affect the Maximum Power Point Tracker (MPPT) operation, however, for the purpose of this project, the linearization was implemented in firmware to improve the accuracy of the measurements made during the testing phase.

In Figure 6.2 the error of the voltage readings is represented. Since the error is linear, the trend line is also represented, which was used to correct the results and improve the accuracy of the measurements. The results after subtracting the linear error are also represented in the figure, and it can be seen that the error is significantly reduced. The average error reduced to 28.325 mV and 29.825 mV for battery and solar panel respectively.

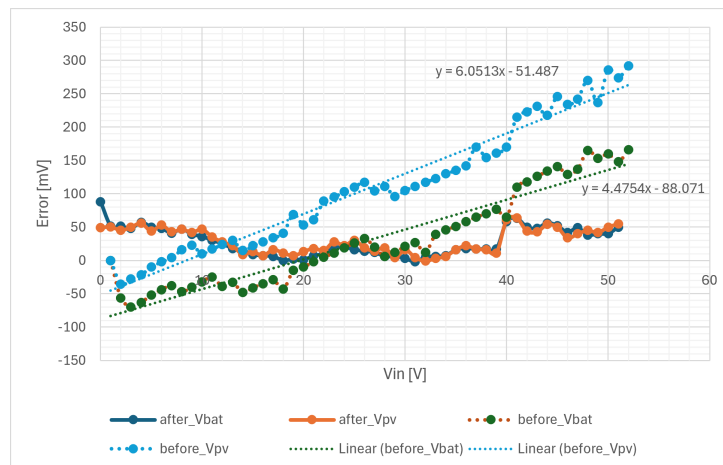


Figure 6.2: Voltage sensor error over voltage.

The same test was performed for the current sensors, the average error was 99.93 mA and 8.5 mA for battery and solar panel respectively, which is acceptable for this application. The error of the current sensors is shown in Figure 6.3, and it can be seen that the error is linear for the battery side and non-linear for the solar panel side. This can be explained by the fact that at low currents, the error is smaller and probably the shunt resistor value in the solar panel side is more accurate than the shunt resistor in the battery side. However, it is still small and does not affect the MPPT operation.

As in the case of the voltage sensor, the subtraction of the linear error would highly increase the accuracy of the measurements. However, it takes a lot of time to do the measurements and calculate the linearization for each PCB which is not realistic for a student prototype project like Técnico Solar Boat (TSB). However, to increase the accuracy of the laboratory results, the error correction was implemented to the battery side current sensor. The results after subtracting the linear error are also represented in the figure, and it can be seen that the error is significantly reduced. The average error reduced to 5.28 mA.

6.3 Boost

The Boost test is essential to identify the efficiency and the correct operation of the synchronous boost converter without the influence of the control system. Additionally, it is essential to confirm the correct di-

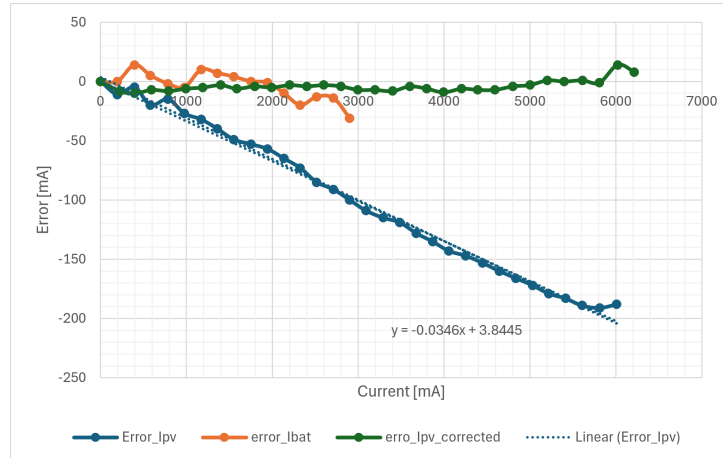


Figure 6.3: Current sensor error over current.

mentioning of the power converter enabling a comparison with simulations and theoretical calculations.

The test was performed using a voltage source on the input side and two power resistors on the load side. Both duty cycle and load resistance were varied to obtain the results in all different points of the charging curve, Figure 5.2.

All the measurements were performed with multimeter with high precision. The voltage readings were performed as close as possible to the PCB, as for the current readings, they have to be performed in line.

It is important to note that this board uses the battery side to power itself, however in this test the battery side is not connected to a battery but since the Photovoltaic (PV) side is connected to a voltage source, this energy flows to the battery side through the boost converter diodes/body diodes and therefore powering the control system.

6.3.1 Switching noise

This subsection will be rewritten with results from V1.2. The idea is to characterize the noise with a spectrum analyzer and compare with CISPR 25 regulation (regulations for conductive EMI measurement).

During the initial experimental tests, a significant amount of electrical noise was observed in all measured signals. The measured noise caused the current to exceed the configured Over Current (OC) protection thresholds, resulting in repeated activation of the protection mechanism. Initially, it was assumed that the problem originated from the PCB measurement system. Consequently, several modifications were implemented in the firmware, mainly increasing the ADC sampling time and increasing the number of samples acquired for each measurement, with the final measurement being obtained by averaging the collected samples. Furthermore, the fault handling logic was modified, because once a OC fault occurred, the converter entered an oscillatory state in which it repeatedly enabled and disabled

itself. Now when a fault occurred, it waits 1 second and verifies if the fault was removed. Although these software modifications improved the stability and accuracy of the measurements, they did not eliminate the excessive noise.

During the OC events, a large oscillation in the input current was also observed, varying approximately between 4 A and 12 A. One possible explanation considered was that the laboratory power supply was operating close to its current limit, causing it to become unstable. Replacing the power supply was therefore considered as a potential solution. However, since the observed noise was considerably higher than expected, it was believed that the root cause was located elsewhere.

A detailed analysis of the reference evaluation board revealed a significant difference in the output decoupling network. While the initial prototype employed a single 1 μ F ceramic capacitor, which completely eliminated the switching noise in simulation, the reference evaluation board used ten 1 μ F ceramic capacitors connected in parallel.

To evaluate the impact of the output capacitance, the converter was characterized without any changes. Figure 6.4 shows the initial noise.

In all oscilloscope captures, Channel 1 (yellow) corresponds to the output current (I_{out}), Channel 2 (blue) to the input voltage (V_{in}), Channel 3 (orange) to the input current (I_{in}), and Channel 4 (green) to the output voltage (V_{out}). Although the screenshots are not of optimal quality, they clearly illustrate the reduction in switching noise obtained after increasing the output decoupling capacitance.

To increase the local output decoupling, nine additional 1 μ F ceramic capacitors were soldered in parallel with the existing capacitor, matching the total capacitance used in the evaluation board. After this modification, a significant reduction in the measured noise was observed, as shown in Figure 6.5.

These results confirm that the insufficient output decoupling capacitance was the primary contributor to the observed electrical noise. Increasing the number of ceramic capacitors substantially improved the converter performance and reduced measurement fluctuations.

Although the results were already satisfactory, the reference evaluation board also includes seven additional 220 nF ceramic capacitors distributed across the output stage. Initially the board also had one smaller capacitor (100 nF) and since it was not possible to get 220 nF with 100 V rating, a six 100 nF capacitors were added. The result should be similar. These capacitors are expected to further reduce the high frequency switching noise. The results are presented in Figure 6.6.

The approximate peak noise amplitudes measured for each hardware configuration are summarized in Table 6.1. Adding nine 1 μ F ceramic capacitors significantly reduced both the current and voltage noise. Subsequently, the six additional 100 nF ceramic capacitors further reduced the current noise, although the input voltage increased but it may be due to imprecise measurements from the oscilloscope.

x

Table 6.1: Approximate noise amplitudes measured for the three hardware configurations. I do not like how these values were measured. In reality the values are less than a quarter. They must be measured separately with a good trigger. I may redo it in V1.3.

Quantity	Original	+9 × 1 μ F	+9 × 1 μ F +6 × 100 nF	...+ schottky diode
I_{in}	1.8 A	640 mA	292 mA	140 mA
I_{out}	0.8 A	300 mA	220 mA	272 mA
V_{in}	2.0 V	350 mV	380 mV	490 mV
V_{out}	8.0 V	2.3 V	2.2 V	1.440 V

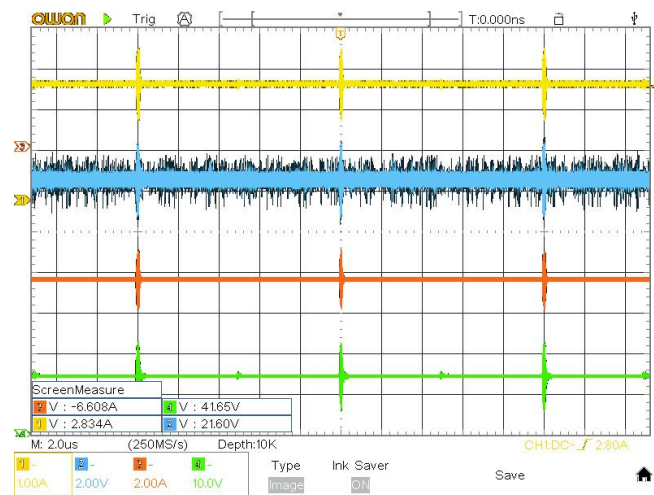


Figure 6.4: Measured waveforms before adding the additional ceramic capacitors.

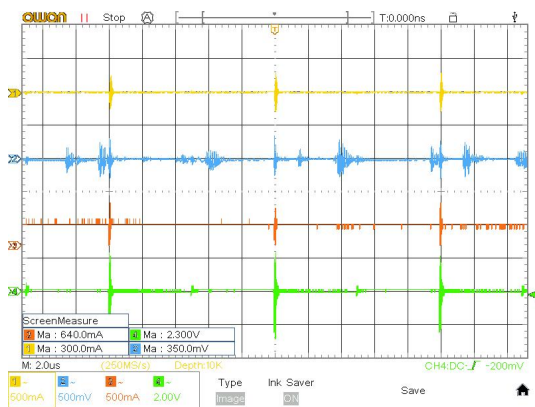


Figure 6.5: Measured waveforms after adding nine additional 1 μ F ceramic capacitors.

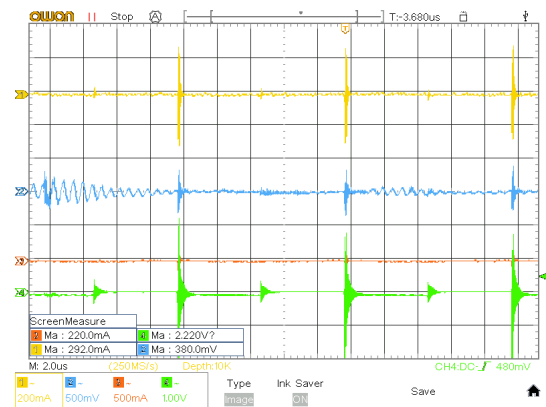


Figure 6.6: Measured waveforms after adding nine additional 1 μ F ceramic capacitors.

6.3.2 Efficiency

The efficiency of the MPPT charger is one of the most important parameters to be measured since it directly affects the power that is effectively extracted from the solar panel and delivered to the battery. As in simulations, the efficiency was measured for each point of the charging curve. All the measurements

were taken with the multimeter and the results are presented in Table 6.2.

Table 6.2: Efficiency measurements for the different test cases.

Case	Multimeter			- OC power	- OC+Transistors power
	Power in	Power out	Efficiency	Efficiency	Efficiency
Case 1	127.444	123.413	96.84%	97.23%	97.64%
Case 2	129.108	124.685	96.57%	96.96%	97.36%
Case 3	127.523	123.232	96.64%	97.02%	97.43%
Case 4	73.1277	70.818	96.84%	97.52%	98.24%
OC	0.5103	0	–	–	–
OC+Transistors	1.04279	0	–	–	–

The measured efficiency is more than 96% for all test cases, which is an extraordinary result. This efficiency takes into account all the losses in the converter, including the measurement system and control system. When compared with the power losses estimation made in section 5.2, the efficiency is around 2% lower than expected. This difference can be explained by the fact that the components used can have slightly different characteristics. Also, the losses caused by the parasitic elements of the PCB were not considered in the estimation.

The PCB was also tested in open circuit with and without driving the transistors. The results show the losses of the circuit without the boost converter. Around 0.5 w of losses are caused only by the measurement and control system. When the transistors are driven without any load, the losses of the converter are negligible since the current is too low. So the measured losses are approximately equal to the driving circuit + the measurement and control system. Based on these values, in table 6.2 it is also present a estimate of the efficiency without the measurement and control system and another without the driving circuit too. The converter efficiency is around 97% to 98% for all test cases, which is close to the simulation measurements. From another perspective, it shows just how significant the losses are on the rest of the circuit.

6.3.3 Temperature rise

The temperature of each element was measured before each test and after 10 min. All test show similar result, a temperature rise of up to 10°C. The inductor is the main heat element, reaching 43°C, the capacitors also reach around 40°C however they are placed side by side with the inductor which heats the capacitors. As expected, the transistors do not overheat. The maximum temperature measured was 39.5°C.

The results are close to what was expected. In Section 5.2, we concluded that the component with the highest losses is the inductor, followed by the transistors and the capacitors. In this section, it is expected that the temperature follows the same order since it is related to the losses. However, the measurements do not follow the expected results. This is explained by the fact that each of these

components have different surface areas, and therefore their heat dissipation is not comparable.

Thermal images of the circuit were also obtained, as shown in Figures 6.7 and 6.8, allowing the areas with the highest temperatures to be identified. However, the quality of the images is limited, and a vertical offset can be observed between the thermal and visible images. Furthermore, the vertical lines present in the thermal images are measurement artefacts caused by the thermal imaging device and do not represent actual temperature variations.

Despite these limitations, the results indicate that the inductor is the component reaching the highest temperature, with a noticeable spread of heat to the surrounding area. The capacitors also exhibit some heat dissipation, although at a lower magnitude than the inductor. On the rear side of the PCB, the only components showing a significant increase in temperature are the transistors.

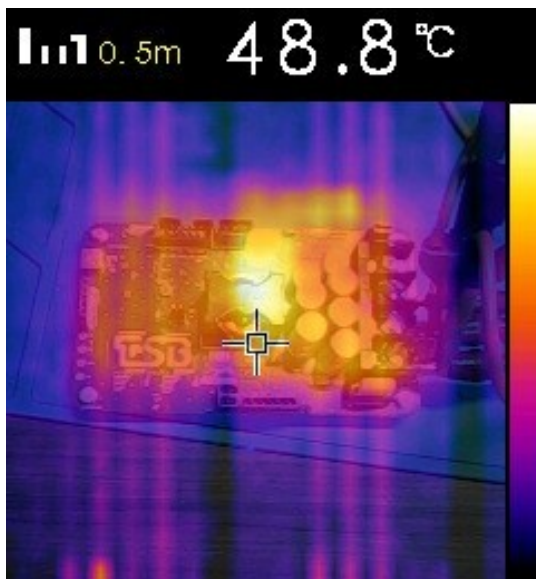


Figure 6.7: Front side thermal image.

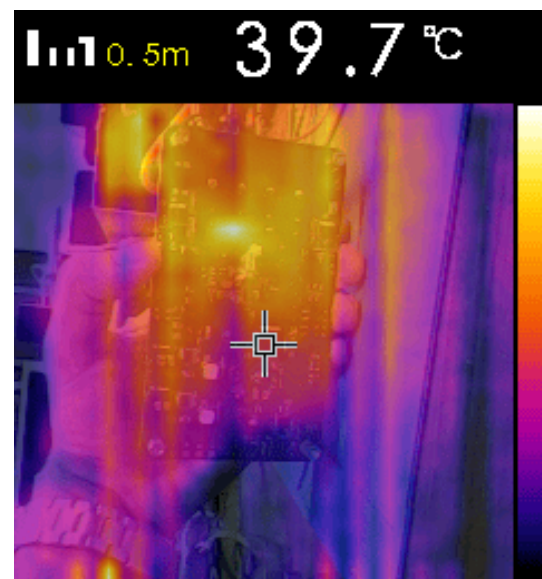


Figure 6.8: Back side Thermal image.

To confirm that the temperature does not rise to a concerning level, a test of 3 hours was performed with the converter operation at Case 3. In figure 6.10 it is represented the temperature evolution over the 3 hours for the inductor, capacitors, transistors and the ambient temperature. These results were acquired using the thermistors placed on the board which measure the ambient temperature and the transistors temperature. The other measurements were acquired using additional external thermistors that can be read by the board using the connector placed for thermistors.

The measured values were confirmed using an infrared thermometer which shows insignificant different temperature, confirming the quality of the measurements.

From Figure 6.10, it can be concluded that after 40 min at maximum power, all the temperatures stop rising. Some oscillations were measured during the experiment due to the fact that ambient temperature also oscillated. As seen earlier the inductor is the element which heats the most, followed by the

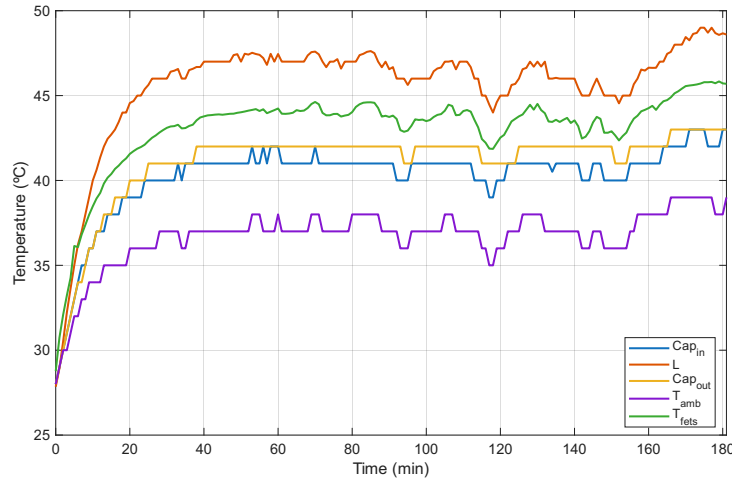


Figure 6.9: Temperature evolution during 3 hours.

transistor and capacitors. After 3 hours, none of the elements reached a concerning temperature.

With this test it is concluded that there is no need for cooling. At least in the same conditions of the test. To confirm this statement, the board was placed inside a cardboard box since TSB usually places boards inside close boxes. If the components inside the box need cooling, a fan or a water cooling needs to be added.

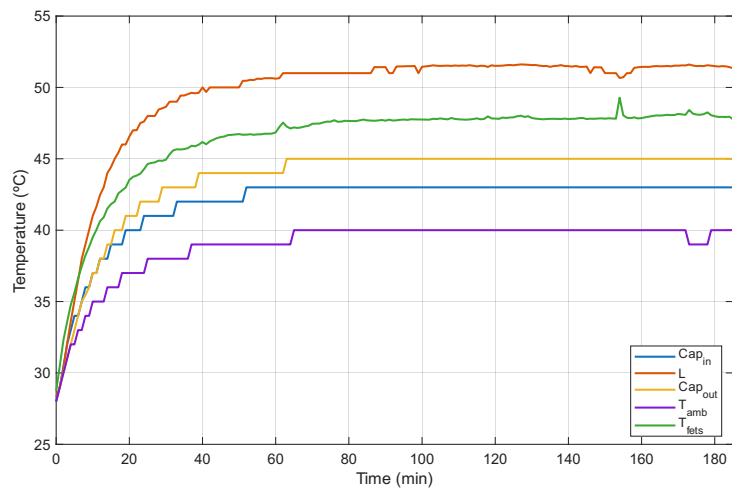


Figure 6.10: Temperature evolution during 3 hours in a close box.

In Figure 6.10 is represented the described test. Since there is no air flow, the ambient temperature reading is more stable and therefore the others too. As in the previous test, after 60 minutes the temperatures remained approximately constant and the maximum temperature was 51.6°C. Since none of the temperatures show concerning values, it is safe to conclude that no additional cooling is needed

6.4 MPPT

In this section the charging state machine is tested as well as the MPPT algorithm to ensure the functionality of the prototype.

In the following tests a laboratory setup was used, where a power supply (the SM 1500-CP-30) was used with an internal function which can simulate the behavior of a solar panel. This function was configured as a 35 cells solar panel at 25°C and variable irradiance controlled by sending TCP messages.

The load was the battery from São Rafael 03 (SR03), which is described in section 2.3.

This setup ensures a controlled environment where the solar panel works at a specified temperature and irradiance parameters increasing the number of controlled variables. Since the environment conditions are set by the user, this setup can be easily replicated for future comparisons and therefore increasing the scientific value of the tests.

6.4.1 Efficiency

To validate the efficiency of the converter over different current and voltage inputs, an irradiance sweep was performed with 10 w/m^2 steps from 100 w/m^2 to 1000 w/m^2 . Each irradiance level is maintained over 2 seconds to ensure that the algorithm is stable. All the values measured during this time are used to perform an average. Additionally, outliers were removed for better results.

The result is an efficiency curve (Figure 6.11) that increases with both the input current and the irradiance, which are approximately proportional. Since the test was performed with different solar panel parameters, three curves were obtained. The converter is clearly more efficient at higher input voltages, that is, for panels with more cells in series.

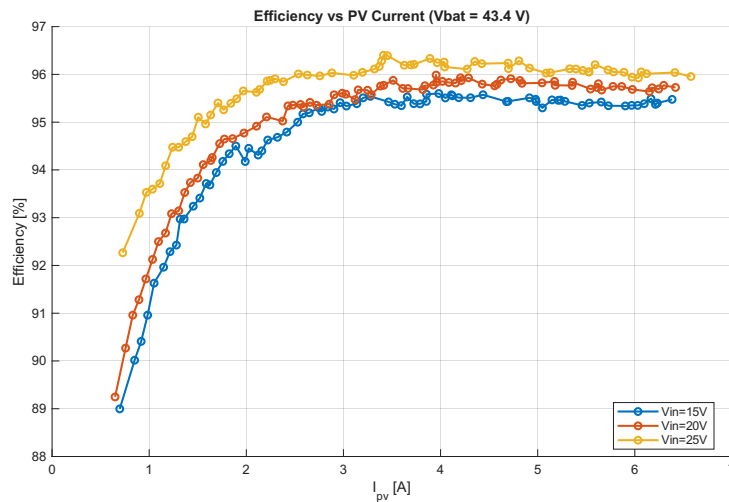


Figure 6.11: Converter efficiency over different input currents and voltages.

Both trends are expected. At low currents the efficiency is lower because the constant losses required to power the electronics (MCU, sensors, etc.) have a larger relative impact. As the current increases, those losses become less significant and the efficiency stabilizes. After around 3 A, the converter switching losses become dominant. Since they increase with the input current, a slight downward slope of the curve is expected.

The higher efficiency at higher input voltage is explained by the fact that a boost converter is more efficient at lower duty cycles. From equation 4.2, for a given output voltage the duty cycle decreases as the input voltage increases. Therefore, increasing the number of cells in series reduces the duty cycle and improves the efficiency.

On the other hand, if the input voltage is constant and the output voltage increases (Figure 6.12), the efficiency will reduce due to the same reason. Equation 4.2 tells us that for a fix input voltage the duty cycle will increase and therefore the efficiency will decrease.

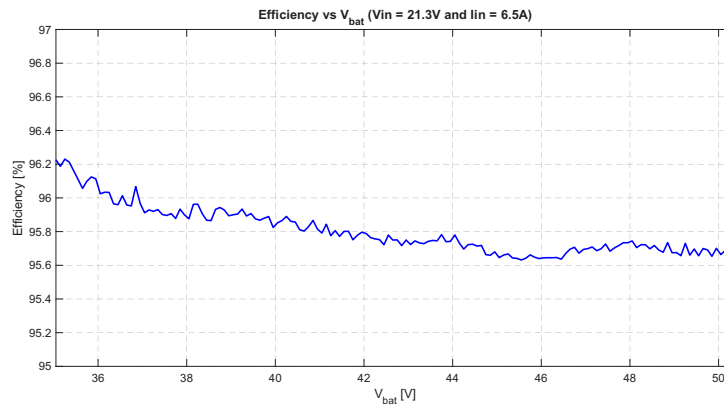


Figure 6.12: Converter efficiency over output voltage.

6.4.2 Tracking speed

Several configurations were tested to optimise the Perturb and Observe (P&O) algorithm. Following the tuning process, the final configuration was defined as shown in Table 6.3.

Table 6.3: MPPT parameters.

Sampling frequency	MPPT frequency	Step size	Power Dead Band
1 kHz	50 Hz	0.7 %	700 mW

To evaluate the performance of the MPPT algorithm under varying environmental conditions, several Python scripts were developed to control the temperature and irradiance of the power source. Through a LAN connection, the scripts transmit TCP messages to the power supply, allowing its environmental parameters to be modified during the tests.

Two test scenarios were defined:

1. **Irradiance staircase:** The test starts at an irradiance of 1000 W/m^2 , which is subsequently reduced by 200 W/m^2 at approximately 2 s intervals until reaching 200 W/m^2 . The irradiance is then increased using the same sequence until returning to 1000 W/m^2 .
2. **Temperature staircase:** The test starts at 25°C , after which the temperature is increased in 10°C steps at 2 s intervals until reaching 85°C . The temperature is then decreased using the same sequence until returning to 25°C .

Both tests were divided into two phases. In the first phase, the environmental parameters were varied gradually to evaluate the tracking performance under normal operating conditions. In the second phase, more abrupt changes were introduced to assess the tracking performance under more demanding conditions. This approach allows the response of the algorithm to be evaluated for both gradual and rapid variations in the operating conditions.

The results of the irradiance and temperature staircase tests are presented in Figures 6.13 and 6.14, respectively. Overall, the algorithm demonstrated satisfactory tracking performance, maintaining the operating point close to the Maximum Power Point (MPP) while responding to changes in the environmental conditions.

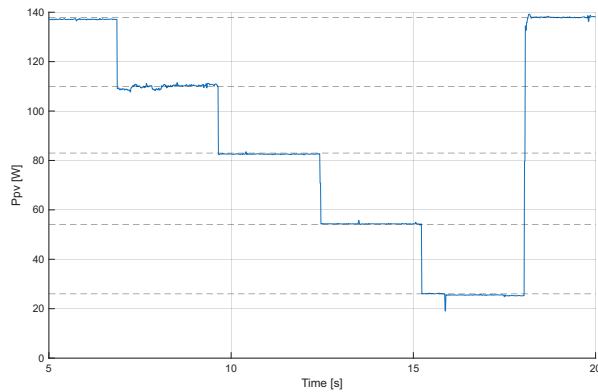


Figure 6.13: Power response for the "Irradiance staircase" test.

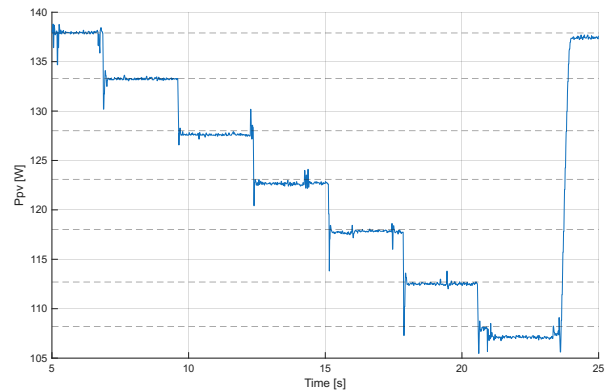


Figure 6.14: Power response for the "Temperature staircase" test.

For both tests, the measured tracking time was typically below 100 ms, based on the measurements obtained during the first phase. When the irradiance was increased from 200 to 1000 W/m^2 , the tracking time was slightly higher but remained below 100 ms. The highest measured tracking time occurred when the solar panels were cooled from 85°C to 25°C , for which a tracking time of approximately 200 ms was measured. Individual tracking times for each test condition are not presented, as the response was nearly instantaneous in most cases, making accurate measurement difficult with the available instrumentation.

In addition to the tracking speed, the MPPT algorithm demonstrated a high tracking accuracy, with an average value of approximately 99%. Nevertheless, some deviations from the MPP were observed under

specific operating conditions, particularly at 85 °C. These deviations are likely related to measurement noise, although this could not be conclusively verified.

6.5 Field test

To prove the functionality of the developed prototype, two charging tests were recorded. Both tests use the smaller battery described in chapter 5.3.

By using the SM 1500-CP-30 power supply in photovoltaic function connected to the MPPT charger, the battery was fully charged. Figure 6.15 shows the result where the CC-CV charging curve was accomplished. In Constant Current (CC) mode, the algorithm tracked the MPP successfully. At Constant Voltage (CV), the output voltage is constant, and the output current dropped slowly.

O fim do teste está miserável, era bom tentar corrigir isto e mostrar que o conversor entra em charge end

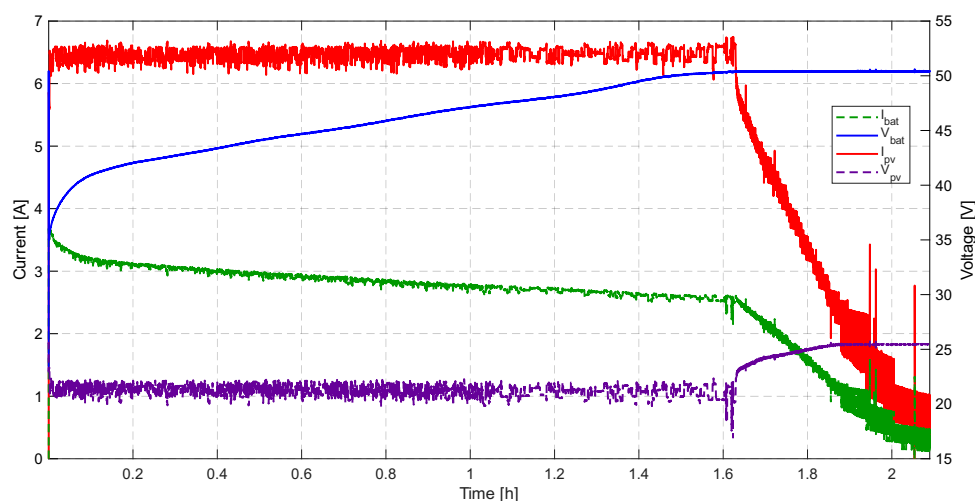


Figure 6.15: Full battery charge in laboratory.

Parei o teste duas vezes e é facilmente notavel no grafico. Em cerca de 3h 30min tive de mudar de sitio pq fiquei sem sol. Perto das 6h percebi que os paineis não estavam a operar em MPP devido a uma limitação do duty cycle maximo que aconteceu pela primeira vez devido ao painel ser muito pequeno.

The last test is a field test with a solar panel of 20 cells submitted to the environment conditions. The test was stopped two times. The first one around 3.5h to change the test spot and the second around 5.45h to correct a mistake. At first the algorithm was never reaching MPP, i though that was due to the solar panel temperature and therefore the solar panel was cooled with water several times during the test. Almost at the end, i realized that the duty cycle was reaching the limit imposed in firmware and therefore it was never reaching MPP. This was never a problem since i had not tested the algorithm with such small solar panel. The rest of the test went as expected.

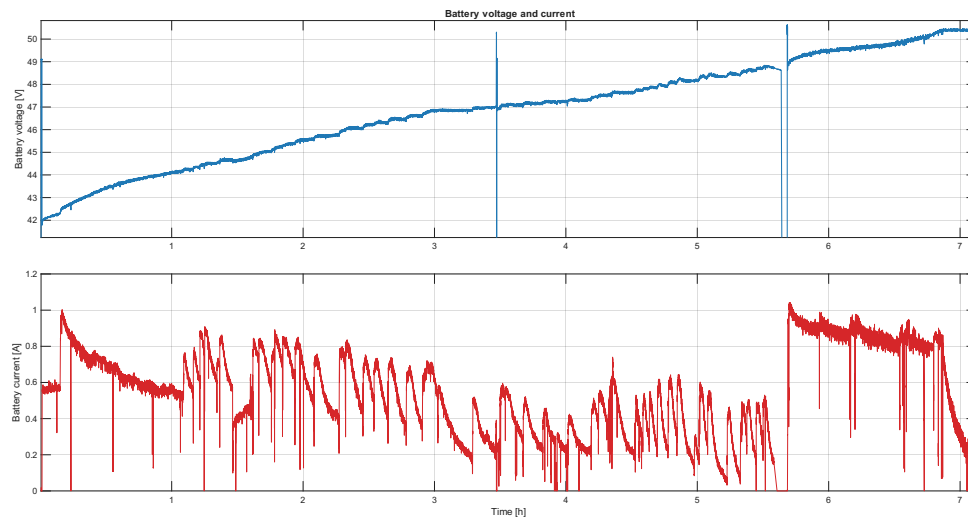


Figure 6.16: Battery charging test. Voltage and current curve.

Show the MPPT in difrent states. No load-Sweep-CC-CV-no load-CV-CC

Repetir teste.

Se possivel adicionar testes em água.

7

Conclusion

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last thing to write

7.1 Future Work

The prototype developed in this thesis provides a functional platform for Técnico Solar Boat (TSB), but several improvements remain possible.

On the control side, additional Maximum Power Point Tracker (MPPT) algorithms should be implemented, tuned and compared under the same operating conditions. A dedicated irradiance measurement, used together with the existing temperature sensing, would also enable a more accurate estimation of conversion losses and could serve as an input to more advanced tracking methods.

Regarding the processing unit, a Microcontroller Unit (MCU) with a native 16-bit ADC would improve measurement resolution. A higher clock frequency would increase the PWM duty cycle resolution, which is particularly relevant for variable step size algorithms.

Finally, both cost and output voltage ripple could be reduced by coordinating the MPPTs that operate in parallel. If the converters are interleaved, that is, operated with a controlled phase shift, the individual output ripples tend to cancel rather than add. A common output capacitor bank could then be shared among chargers, reducing the required capacitance per unit and the overall cost of the installation.

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